DESIGN AND SIMULATION OF FIELD ORIENTED CONTROL FOR PMSM DRIVES AND ITS APPLICATION IN ELECTRIC BIKE MODELLING AND ANALYSIS

Project Report

Submitted in partial fulfilment of the requirements for the award of the degree of

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in

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by

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CERTIFICATE

This is to certify that the report entitled “Design and simulation of field-oriented control for PMSM drives and its application in electric bike modelling and analysis” is a bona fide record of the project presented by Amitha Thomas, B Harisanth Krishna, Gautham Krishna M and Gowri Priya Vinu, in partial fulfilment of the requirements for the award of the degree of Bachelor of Technology in Electrical and Electronics Engineering from APJ Abdul Kalam Technological University.

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We hereby declare that this submission is our own work and that, to the best of our knowledge and belief, it contains no material previously written by another person nor material which has been accepted for the award of any other degree or diploma of the university or other institute of higher learning, except where due acknowledgement has been made in the text.

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ABSTRACT

This report presents the comprehensive design and engineering analysis of a full-size electric motorcycle developed through the retrofitting of an NS160-class chassis, targeting a top speed of 70–80 km/h. The powertrain is built around a 2000 W Permanent Magnet Synchronous Motor (PMSM) operating at a nominal voltage of 72 V, controlled by a FarDriver ND72340 motor controller employing Field-Oriented Control (FOC) for precise torque and speed regulation. A 72 V, 30 Ah lithium-ion battery pack configured in a 20S6P arrangement provides a total energy capacity of approximately 2.16 kWh, managed by a DALY BMS rated at 100 A continuous discharge. Power is transmitted to the rear wheel via a chain drive with a 5.14:1 gear ratio, delivering a wheel torque of approximately 210 N·m and a cruising speed of 65 km/h at an overall system efficiency of approximately 79 %.

The structural and dynamic systems were designed with an emphasis on safety, stability, and rider comfort. The chassis, based on the NS160 frame, was validated through ANSYS finite element analysis, revealing a maximum von Mises stress of 66.812 MPa against a factor of safety exceeding 3. The suspension system employs an upside-down telescopic front fork and a Nitrox mono-shock rear absorber, with calculated spring rates of 22 N/mm (front) and 29.4 N/mm (rear), a rear natural frequency of 1.7 Hz within the comfort range, and a mono-shock FEA factor of safety between 15 and 18. The hydraulic disc braking system demonstrated an average stopping distance of 2.54 m and an average braking time of 0.41 s, with thermal analysis confirming acceptable heat dissipation under simulated braking conditions.

The steering geometry, with a measured rake angle of 26.5° and a wheelbase of 1372 mm, yields a turning radius of approximately 1959 mm, ensuring stable straight-line performance and adequate low-speed maneuverability. The electrical architecture integrates a buck converter (72 V to 12 V/5 V), a pre-charge circuit, ANL and MCB overcurrent protection, and a 12 V relay-based HV control system, all wired in compliance with competition safety standards. The integrated design demonstrates that the retrofitted electric motorcycle satisfies the structural integrity, dynamic performance, energy efficiency, and safety requirements,establishing it as a viable and practical electric two-wheeler platform.

CO – PO / PSO ARTICULATION MATRIX

Course Outcome	Program Outcome (PO)											PSO		
	1	2	3	4	5	6	7	8	9	10	11	1	2	3
CO1: Model and solve real world problems by applying knowledge across domains	2	2	2	1	2	2	1	1	1	1	1	2	3	2
CO2: Develop products, processes or technologies for sustainable and socially relevant applications	2	2	2		1	3	1	1		1	1	2	3	2
CO3: Function effectively as an individual and as a leader in diverse teams and to comprehend and execute designated tasks							3	2	2	1	2	2	3	2
CO4: Plan and execute tasks utilizing available resources within timelines, following ethical and professional norms					2		3	2	2	3	2	2	3	2
CO5: Identify technology/research gaps and propose innovative/creative solutions	2	3	3	1	2					1		2	3	2
CO6: Organize and communicate technical and scientific findings effectively in written and oral forms					2		2	2	3	1	1	2	3	2

SUSTAINABLE DEVELOPMENT GOALS (SDG) ADDRESSED

SDG	Target	Justification
SDG 7	7.2: Increase substantially the share of renewable energy in the global energy mix	The electric bike operates on electrical energy, enabling integration with renewable energy sources and reducing dependence on fossil fuels.
SDG 7	7.3: Double the global rate of improvement in energy efficiency	The use of Field-Oriented Control (FOC) ensures high motor efficiency and optimal energy utilization from the battery.
SDG 9	9.4: Upgrade infrastructure and retrofit industries to make them sustainable, with increased resource-use efficiency and greater adoption of clean and environmentally sound technologies	The project demonstrates adoption of clean and efficient electric mobility technology, contributing to sustainable transport infrastructure development.
SDG 9	9.5: Enhance scientific research, upgrade the technological capabilities of industrial sectors in all countries	Development of FOC-based PMSM control and EV integration reflects advanced engineering research and technological innovation.
SDG 11	11.2: safe, affordable, accessible and sustainable transport systems	The electric bike provides a low-emission, energy-efficient alternative for urban transportation.
SDG 11	11.6: Adverse per capita environmental impact of cities and special attention to air quality and municipal and other waste management	Electric propulsion eliminates tailpipe emissions, reducing urban air pollution and noise levels.
SDG 12	12.2: Achieve the sustainable management and efficient use of natural resources	Efficient motor control and energy management reduce overall energy consumption compared to conventional vehicles.
SDG 13	13.2: Integrate climate change measures into national policies	The shift from internal combustion to electric mobility directly contributes to reduction of greenhouse gas emissions.

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AC	Alternating Current
ANL	Automotive Newville Limitless (Fuse Type)
ANSYS	Analysis System (FEA Software)
BMS	Battery Management System
BLDC	Brushless Direct Current (Motor)
CAN	Controller Area Network
CCCV	Constant Current – Constant Voltage
DC	Direct Current
DB	Distribution Bus
DIO	Digital Input Output
DP	Double Pole
DTC	Direct Torque Control
EV	Electric Vehicle
FEA	Finite Element Analysis
FOC	Field-Oriented Control
HV	High Voltage
IGBT	Insulated Gate Bipolar Transistor
IoT	Internet of Things
LV	Low Voltage
MCB	Miniature Circuit Breaker
MOSFET	Metal Oxide Semiconductor Field Effect Transistor
NTC	Negative Temperature Coefficient
PI	Proportional–Integral
PMSM	Permanent Magnet Synchronous Motor
PWM	Pulse Width Modulation
RFOC	Rotor Field-Oriented Control
RPM	Revolutions Per Minute
RS-485	Recommended Standard 485 (Serial Communication)
SB50	Anderson SB50 Connector
SFOC	Stator Field-Oriented Control
SoC	State of Charge
SVPWM	Space Vector Pulse Width Modulation
UART	Universal Asynchronous Receiver Transmitter
USD	Upside-Down (Telescopic Fork)
V/f	Voltage-to-Frequency (Scalar Control)
VSI	Voltage Source Inverter
UUV	Unmanned Underwater Vehicle

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CHAPTER 1

INTRODUCTION

1.1 Overview

The growing demand for sustainable and energy-efficient transportation has accelerated the shift from conventional internal combustion engine vehicles to electric vehicles (EVs). Factors such as rising fuel costs, environmental concerns, and strict emission regulations have made electric mobility an essential solution. Among EVs, electric bikes have gained popularity due to their low cost, compact design, and suitability for urban transportation

E-bikes utilize electrical energy stored in batteries to drive an electric motor, eliminating the need for fossil fuels. The performance of an E-bike largely depends on its motor drive system. Permanent Magnet Synchronous Motors (PMSMs) are widely preferred in electric vehicles due to their high efficiency, compact size, and superior torque characteristics

However, controlling PMSMs is complex due to their nonlinear and coupled behavior. Conventional control methods such as V/f control provide limited performance and are not suitable for applications requiring precise torque and speed control. To address these limitations, advanced control techniques like Field-Oriented Control (FOC) are used. FOC enables independent control of torque and flux by transforming three-phase currents into a rotating reference frame, resulting in improved efficiency, smooth operation, and fast dynamic response.

Overall, the project demonstrates a reliable, efficient, and scalable electric bike system with validated performance, smooth control characteristics, and readiness for final integration and real-world testing.

1.2 Market values

The developed electric dirt bike, based on a 2 kW PMSM motor and a 72 V, 30 Ah lithium-ion battery, can be positioned competitively in the Indian electric two-wheeler market within the 1.5–1.8 lakh price range. With an approximate weight of 90 kg, the vehicle offers a superior power-to-weight ratio compared to many commercial counterparts in the same segment, resulting in better acceleration and efficient energy utilization. The integration of Field-Oriented Control (FOC) enables precise torque and speed control, enhancing overall performance and ride smoothness

In comparison to existing market options with similar power ratings, this bike can be considered one of the fastest in its category, with an expected top speed of around 70-80 km/h, depending on tuning and load conditions. The system's performance is further strengthened by its high torque capability, optimized drivetrain, and lightweight design, making it suitable for both urban commuting and light off-road applications.

Additionally, the implementation of layered protection systems, including BMS, fuse, MCB, contactor, and a pre-charge circuit, ensures safety, reliability, and durability, which are critical factors for market acceptance. Overall, the project demonstrates a cost-effective, high-performance electric bike solution that combines advanced control strategies with practical engineering design, positioning it as a strong competitor in the entry-level performance EV segment.

1.3 Project Objective

This project aims to design, develop, and validate a full-size retrofitted electric motorcycle based on the Bajaj NS160 chassis platform, capable of achieving a target top speed of 65–70 km/h. The powertrain integrates a 2000W Permanent Magnet Synchronous Motor (PMSM) driven by a 72V/30Ah lithium-ion battery pack, managed through a FarDriver ND72340 Field-Oriented Control (FOC) motor controller, and transmitted to the rear wheel via a 5.14:1 chain drive system. The project seeks to demonstrate that electric propulsion can be seamlessly retrofitted into a conventional motorcycle platform without compromising the ride feel, ergonomics, or structural behaviour of the original vehicle.

The engineering scope encompasses the complete analysis and validation of all principal subsystems — chassis, steering, suspension, braking, transmission, motor control, and electrical architecture. Structural integrity of the NS160 frame has been verified through FEA, confirming a maximum stress of 66.812 MPa and a factor of safety exceeding 3 under a combined rider and component load of 2207 N. The suspen-

sion system employs a USD telescopic front fork and a Nitrox mono-shock rear unit, tuned to a rear spring rate of 29.4 N/mm and a natural frequency of 1.7 Hz, ensuring ride comfort within the optimal range. A hydraulic disc brake system on both axles delivers an average stopping distance of 2.54 m, meeting competition safety requirements.

The electrical and power system architecture has been designed with layered protection and efficiency in mind. The battery pack, configured in a 20S×6P arrangement of lithium-ion cells, delivers a nominal voltage of 72V with a total energy capacity within the permissible 1.5–2.5 kWh competition window, protected by a Daly BMS rated at 100A continuous discharge. The HV architecture incorporates an ANL fuse, DC-DP MCB, pre-charge circuit, and a 72V-to-12V buck converter to safely power auxiliary systems via Arduino-based control. With an overall system efficiency of approximately 79 % and a wheel torque output of 210 N·m, the project establishes a technically rigorous, competition-ready electric motorcycle that balances performance, safety, and practical rideability.

1.4 Methodology

1.4.1 System Modelling

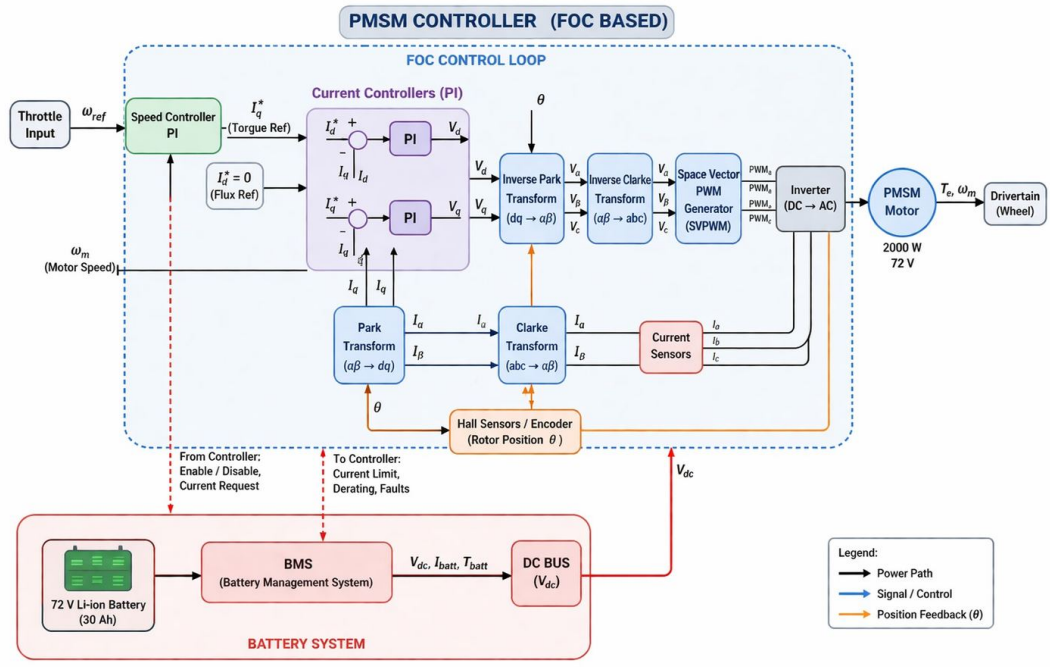


Figure 1.1: Block Diagram of FOC for PMSM

The control strategy implemented for the PMSM drive system is Field-Oriented Control (FOC). The core idea of FOC is to decompose the three-phase stator currents into two orthogonal components in a rotating reference frame — the direct-axis current I_d, controlling magnetic flux, and the quadrature-axis current I_q, controlling electromagnetic torque. This decoupling allows independent and precise control of torque and speed, making the PMSM behave similarly to a separately excited DC machine. The complete closed-loop system is illustrated in Fig. 1.1.

Forward Control Path

The reference direct-axis current is set to zero ($I_d^* = 0$) for maximum torque per ampere operation, while the reference quadrature-axis current I_q^* is derived from the throttle input. Two PI controllers regulate the d-axis and q-axis current errors independently, producing voltage commands V_d and V_q as given by:

$$V_d = K_p \cdot e_d + K_i \int e_d \, dt, \qquad V_q = K_p \cdot e_q + K_i \int e_q \, dt \tag{1.1}$$

These commands are then converted from the rotating d-q frame to the stationary α-β frame via the Inverse Park Transform, and subsequently expanded into three-phase voltage references V_u, V_v, V_w using the Inverse Clarke Transform. These references generate PWM signals that drive the six-switch inverter of the FarDriver ND72340, producing the three-phase AC supply to the PMSM stator.

Feedback Path and Rotor Position Sensing

The measured three-phase stator currents I_u, I_v, I_w are transformed back into the d-q frame through the Clarke and Park transforms in sequence, using the rotor electrical angle θ:

$$I_d = I_\alpha \cos\theta + I_\beta \sin\theta, \qquad I_q = -I_\alpha \sin\theta + I_\beta \cos\theta \tag{1.2}$$

The resulting I_d and I_q values are fed back to the PI controllers, closing the control loop. The rotor angle θ is continuously estimated by the FarDriver ND72340 from the Hall sensor commutation signals, which produce logic-level transitions at every 60° electrical interval, ensuring accurate synchronisation of all coordinate transformations throughout operation.

1.4.2 Electric Bike Overview

Vehicle Platform and Chassis

The electric motorcycle is developed on a retrofitted Bajaj NS160 chassis, targeting a top speed of 70–80 km/h. The frame retains its original geometry with a wheelbase of 1372 mm and a rake angle of 26.5°, ensuring the gyroscopic and handling characteristics of the base platform are preserved. The total vehicle mass including the rider is approximately 225 kg, with a centre of gravity height of 805 mm. Front suspension is a USD telescopic fork and the rear uses a Nitrox mono-shock absorber, with a rear spring rate of 29.4 N/mm tuned for a natural frequency of 1.7 Hz within the comfort range. Hydraulic disc brakes are fitted on both wheels, achieving an average stopping distance of 2.54 m from operating speed.

Propulsion System

The traction motor is a QS Motor 2000 W inner-rotor PMSM with a rated voltage of 72 V and a rated speed of 2800 RPM, extendable to 4400 RPM under flux weakening. The motor produces a maximum torque of 45.4 N·m and operates at a peak efficiency of 88%. Power is transmitted to the rear wheel through a roller chain drive with a 14-tooth motor sprocket and a 42-tooth rear sprocket, giving a gear ratio of 1:3, which delivers a wheel torque of approximately 210 N·m and a vehicle cruising speed of 65 km/h. The motor is controlled by a FarDriver ND72340 controller operating on Field-Oriented Control (FOC) with Hall sensor feedback, supporting a maximum phase current of 340 A and a line current of 75 A in sport mode.

Battery and Electrical Architecture

The energy storage system consists of a 72 V, 30 Ah lithium-ion battery pack configured in a 20 series and 6 parallel (20S6P) arrangement using 3.6 V, 5000 mAh cylindrical cells, providing a total energy of 2.16 kWh within the competition-mandated range of 1.5–2.5 kWh. A Daly BMS with a 100 A continuous discharge rating provides full cell-level protection including overvoltage, undervoltage, overcurrent, and thermal cutoff. The high-voltage circuit is protected by an 80 A ANL fuse and a DC DP C63 MCB, while an SB50 Anderson connector is used at the battery output. A 72 V to 12 V buck converter supplies the auxiliary low-voltage systems, and a precharge circuit limits inrush current at startup. The bike is charged using an 87 V, 6 A EV charger via constant current–constant voltage (CCCV) charging.

CHAPTER 2

REVIEW OF LITERATURE

2.1 Paper 1: The Principle of Field Orientation Applied to the New Transvector Closed-Loop Control System for Rotating-Field Machines[1]

paper introduced Field-Oriented Control (FOC), a fundamental breakthrough in AC motor control that made it possible to achieve DC motor-like performance in machines such as PMSM and induction motors. The key idea is to transform the three-phase stator currents into a rotating reference frame aligned with the rotor flux using coordinate transformations (later known as Clarke and Park transforms). This allows the current to be separated into two independent components: direct-axis current (Id), which controls the magnetic flux, and quadrature-axis current (Iq), which controls torque. By decoupling these two components, the control system can regulate torque and flux independently using simple PI controllers, resulting in precise, fast, and stable motor operation. Compared to traditional methods like V/f control, this approach significantly improves dynamic response, reduces torque ripple, and increases efficiency. Although the method initially required accurate rotor position information and high computational capability, modern controllers have overcome these limitations, making FOC the standard technique in high-performance drives. This principle is directly applied in PMSM-based electric vehicles, including this project, where it enables smooth acceleration, accurate speed control, and efficient energy utilization.

2.2 Paper 2 : Sensorless Vector and Direct Torque Control[2]

presents advanced control strategies for AC machines, particularly focusing on eliminating the need for mechanical sensors in vector control systems. The work extends the concept of Field-Oriented Control (FOC) by introducing sensorless techniques, where rotor position and speed are estimated using electrical parameters such as voltage and current instead of physical encoders. This is highly beneficial in applications like electric vehicles, where reducing cost, size, and maintenance is critical. The book also compares Vector Control (FOC) and Direct Torque Control (DTC), highlighting that FOC provides smooth and precise control with low torque ripple, while DTC offers faster dynamic response but with higher ripple. Vas explains various estimation methods, including model-based observers and adaptive techniques, to accurately determine rotor position in PMSM drives. These strategies improve system reliability and efficiency while maintaining high performance. The concepts discussed are highly relevant to modern EV motor drives, where sensorless FOC is widely used to achieve compact, efficient, and robust control systems.

2.3 Paper 3: Torque Production in Permanent Magnet Synchronous Motors[3]

The paper provides a fundamental analysis of how torque is generated in PMSMs, forming the theoretical basis for modern motor control techniques. The paper explains that electromagnetic torque in a PMSM is produced due to the interaction between the stator current and the magnetic field of the rotor permanent magnets. It introduces the mathematical torque equation in the dq reference frame, showing that torque is directly proportional to the quadrature-axis current (Iq), while the direct-axis current (Id) mainly influences the flux. This clear separation of torque and flux components supports the implementation of Field-Oriented Control (FOC), where Id is typically maintained at zero for maximum efficiency and Iq is controlled to regulate torque. The paper also discusses factors affecting torque such as magnetic flux linkage, current magnitude, and machine parameters, providing insight into torque ripple and performance optimization. This work is essential for understanding PMSM behavior and is directly applicable to electric vehicle drives, including this project, where efficient and precise torque control is required for smooth operation and high performance.

2.4 Paper 4: Implementation of Field-Oriented Control Algorithm for BLDC Motor Speed Regulation[4]

The paper present a practical implementation of FOC for achieving precise speed control in brushless DC motors, which share similar control principles with PMSMs. The paper focuses on converting three-phase motor currents into the dq reference frame using Clarke and Park transformations, enabling independent control of torque and flux components. By employing PI controllers for regulating Id and Iq currents and generating switching signals through PWM techniques, the system achieves accurate speed tracking with minimal steady-state error. The results demonstrate fast dynamic response, reduced torque ripple, and improved efficiency compared to conventional control methods. This work validates the effectiveness of FOC in real-time motor control applications and directly supports the simulation and implementation carried out in this project, where similar control strategies are used for PMSM-based electric bike drive systems.

2.5 Paper 5: Comparison Study of Rotor Field-Oriented Control and Stator Field-Oriented Control in Permanent Magnet Synchronous Motors[5]

This paper present a detailed theoretical and performance comparison between two FOC strategies based on different reference frames. In Rotor Field-Oriented Control (RFOC), the reference frame is aligned with the rotor flux, enabling clear decoupling of torque and flux, which results in accurate torque control and high efficiency. In contrast, Stator Field-Oriented Control (SFOC) aligns the reference frame with the stator flux, offering simpler implementation but slightly reduced performance in terms of dynamic response and precision. The paper evaluates both methods based on parameters such as torque ripple, speed response, and computational complexity. Results indicate that RFOC provides better dynamic performance and smoother operation, making it more suitable for high-performance applications like electric vehicles, while SFOC may be preferred in simpler or cost-sensitive systems. This comparison is important for understanding different control approaches and supports the selection of rotor-oriented FOC used in this project for achieving precise and efficient PMSM control.

2.6 Paper 6: Field-Oriented Control of BLDC Motors for Unmanned Underwater Vehicle (UUV) Applications[6]

The paper present a real-world implementation of FOC in a demanding environment where precise control and reliability are critical. The paper demonstrates how FOC is applied to a BLDC motor to achieve smooth torque, accurate speed control, and efficient operation under varying load conditions encountered in underwater systems. By using Clarke and Park transformations along with PI controllers, the motor currents are regulated in the dq frame, enabling independent control of torque and flux. The results show improved dynamic response, reduced torque ripple, and stable operation even under disturbances. This work highlights the practical effectiveness of FOC in real-time applications and supports its suitability for electric vehicle systems, including this project, where similar control strategies are used to ensure smooth performance, high efficiency, and reliable operation.

2.7 Paper 7: Fuzzy Logic Controller Implementation for BLDC Drive Speed Control Using FOC Approach and SVPWM Based VSI Technique[7]

The paper present an advanced enhancement to conventional FOC by integrating a fuzzy logic controller and Space Vector Pulse Width Modulation (SVPWM). Instead of traditional PI controllers, the fuzzy logic controller is used to handle system nonlinearities and uncertainties more effectively, resulting in improved speed regulation and robustness under varying load conditions. The use of SVPWM further enhances the inverter performance by providing better DC bus utilization, reduced harmonic distortion, and smoother voltage output compared to conventional PWM techniques. The combined approach leads to faster dynamic response, reduced torque ripple, and improved overall efficiency of the motor drive system. This work demonstrates how intelligent control techniques can significantly enhance FOC performance and is relevant to this project as a potential future improvement for achieving more optimized and adaptive control in electric bike applications.

2.8 Paper 8: Permanent Magnet Synchronous and Brushless DC Motor Drives[8]

The paper provides a comprehensive analysis of the design, modelling, and control of PMSM and BLDC motor drives. The work explains the fundamental operating principles, including electromagnetic torque production, back-EMF characteristics, and dynamic behaviour of these machines. It also presents detailed mathematical models in both stationary and rotating reference frames, which are essential for implementing advanced control strategies such as Field-Oriented Control (FOC). The book discusses various control techniques, including current control, speed control, and sensorless methods, along with practical considerations like inverter design, switching strategies, and efficiency optimization. It serves as a key reference for understanding high-performance motor drives and is highly relevant to electric vehicle applications, including this project, where PMSM-based FOC control is used for achieving efficient, reliable, and precise motor operation.

2.9 Paper 9: Sensorless Control of Induction Motor Drives[9]

The paper presents advanced techniques for eliminating mechanical sensors in motor control systems by estimating rotor position and speed from electrical signals. The paper discusses observer-based and model-based estimation methods that use stator voltage and current to accurately determine motor states, enabling reliable control without encoders. Although focused on induction motors, the concepts are highly applicable to PMSM drives and Field-Oriented Control, particularly in developing sensorless FOC systems. Sensorless control reduces system cost, complexity, and maintenance while improving reliability—making it highly suitable for electric vehicle applications. This work is important for future scope in this project, where implementing sensorless FOC can enhance system robustness, reduce hardware dependency, and achieve efficient and compact electric bike drive systems.

2.10 Paper 10: Design and Development of an Innovative E-Bike[10]

The paper present a comprehensive approach to the design, integration, and optimization of an electric bicycle system. The paper focuses on key subsystems such as the battery pack, electric motor, power electronics, and mechanical drivetrain, emphasizing efficiency, lightweight design, and user performance. It discusses energy consumption, vehicle dynamics, and component selection to achieve optimal range

and performance. The study also highlights the importance of proper integration between electrical and mechanical systems to ensure reliability and safety. This work is directly relevant to this project, as it provides practical insights into electric bike design, including system architecture, component sizing, and performance evaluation, which align with the development of the PMSM-based electric dirt bike.

2.11 Paper 11: Design and Implementation of an Electric Bicycle Using BLDC Motor with Optimized SVPWM Technique[11]

The paper present a modern approach to electric bike development focusing on efficient motor control and power electronics. The paper utilizes a BLDC motor driven by a Voltage Source Inverter (VSI) with Space Vector Pulse Width Modulation (SVPWM), which improves DC bus utilization and reduces harmonic distortion compared to conventional PWM methods. The system demonstrates enhanced speed control, smoother operation, and better energy efficiency under varying load conditions. The study also discusses battery integration, controller design, and real-time performance evaluation of the electric bicycle. This work is highly relevant to the current project, as it highlights modern EV implementation techniques and supports the use of advanced control strategies like FOC and SVPWM for achieving high-performance electric bike operation.

2.12 Paper 12: Design and Development of Electric Two Wheeler[12]

This paper present a detailed framework for the modelling, design, and integration of an electric vehicle system. The paper focuses on key aspects such as battery selection, motor sizing, powertrain configuration, and energy management to achieve optimal performance and efficiency. It discusses vehicle dynamics, load requirements, and drivetrain calculations to ensure proper matching between the motor and mechanical system. Additionally, the study highlights the importance of system-level modelling for predicting performance parameters like speed, torque, and range. This work is highly useful for system modelling in this project, as it provides a structured approach for designing and analyzing an electric bike, including component selection, drivetrain design, and overall system integration.

2.13 Paper 13: Modern Electric, Hybrid Electric and Fuel Cell Vehicles[13]

This paper provide a comprehensive overview of electric vehicle technologies, including powertrain architecture, energy storage systems, motor drives, and control strategies. The book covers different EV configurations, battery technologies, and the role of power electronics in efficient energy conversion and control. It explains the operation of electric motors such as PMSM and their integration with controllers for achieving high performance and efficiency. Additionally, it discusses vehicle dynamics, energy management, and system-level design considerations essential for EV development. This work serves as a core reference for understanding the complete electric vehicle system and is highly relevant to this project, as it supports the design, modelling, and analysis of the electric bike powertrain and control system.

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2.14 Paper 14 : Hybrid Electric Vehicles: Architecture and Motor Drives[14]

The paper provide a detailed analysis of hybrid and electric vehicle architectures, focusing on powertrain configurations, motor drives, and energy management strategies. The work explains different vehicle topologies such as series, parallel, and series-parallel hybrids, along with the role of electric machines like PMSM and induction motors in propulsion systems. It also covers power electronic converters, control techniques, and the interaction between mechanical and electrical subsystems. The book emphasizes motor drive selection, efficiency optimization, and performance analysis under various operating conditions. This reference is highly relevant to the project as it supports the understanding of EV powertrain design, motor drive integration, and control strategies required for developing an efficient electric bike system.

2.15 Paper 15: Modeling, Simulation, and Analysis of Permanent Magnet Motor Drives – Part II: The Brushless DC Motor Drive[15]

The paper present a detailed mathematical model and simulation framework for permanent magnet motor drives, forming a strong foundation for modern motor control techniques. The paper develops dynamic equations of the motor using

electrical and mechanical parameters, enabling accurate analysis of torque, speed, and current characteristics under different operating conditions. It explains the relationship between back-EMF, stator currents, and electromagnetic torque, and demonstrates how these variables can be effectively controlled through power electronic converters. The simulation results validate the model's ability to predict real motor behavior, making it highly useful for controller design and performance evaluation. Although focused on BLDC motors, the modelling approach is closely related to PMSM systems and supports the implementation of advanced control methods like Field-Oriented Control (FOC). This work is directly relevant to this project, as it aids in understanding motor modelling, simulation accuracy, and performance analysis of the electric bike drive system.

2.16 Research Gap Identified

- Most studies on Field-Oriented Control (FOC) focus on theoretical modelling and simulation, with limited real-time implementation in electric vehicle systems.
- Conventional control methods such as PI-based FOC are widely used, but they do not address nonlinearities and parameter variations effectively under dynamic conditions.
- Many works discuss sensorless control techniques, but their practical implementation in low-cost electric bike systems remains limited.
- Existing studies on PMSM drives focus on motor performance, with less emphasis on complete system integration including battery, protection circuits, and drivetrain.
- Research on electric bikes mainly addresses mechanical design or basic motor control, lacking advanced control strategies like FOC combined with real hardware validation.
- Limited work is available on coordinated design of HV architecture, including fuse, MCB, relay, and precharge circuits for safe EV operation.
- Most literature does not include experimental validation under real operating conditions, such as load variation and practical testing.
- There is insufficient focus on low-cost, scalable EV solutions suitable for practical applications and competitions.

CHAPTER 3

MODELLING AND ANALYSIS

3.1 PI Controllers

The Proportional–Integral (PI) controller is an essential component in the Field Oriented Control (FOC) of the Permanent Magnet Synchronous Motor (PMSM). It is used to regulate the d-axis (I_d) and q-axis (I_q) currents, which correspond to flux and torque control respectively.

In the FOC scheme, the reference currents (I_d^* and I_q^*) are compared with the measured currents (I_d and I_q) to generate error signals:

$$e_d = I_d^* - I_d \tag{3.1}$$

$$e_q = I_q^* - I_q \tag{3.2}$$

These error signals are processed using PI controllers to produce the required voltage outputs. The mathematical expression of the PI controller is given by:

$$u(t) = K_p e(t) + K_i \int e(t)\, dt \tag{3.3}$$

In the Laplace domain, the transfer function of the PI controller is:

$$G(s) = K_p + \frac{K_i}{s} \tag{3.4}$$

The proportional gain (K_p) provides a fast response to changes in error, while

the integral gain (K_i) eliminates steady-state error by accumulating past errors. The outputs of the PI controllers are the voltage components (V_d and V_q), which are used for further transformations and applied to the inverter.

In this system, the d-axis current is typically maintained close to zero to achieve efficient operation, while the q-axis current controls the electromagnetic torque of the motor.

3.1.1 Performance of PI Controllers

The effectiveness of the PI controllers can be observed from the simulation results. The no-load speed response shows a smooth rise to the steady-state value without significant overshoot, indicating proper tuning of the controllers.

The electromagnetic torque initially exhibits a high value to accelerate the motor and then settles to a low steady value under no-load conditions. This demonstrates that the PI controllers effectively regulate the current components and maintain stable operation.

Under varying speed conditions, the motor accurately tracks the reference speed with minimal delay. The torque response follows the changes in speed, showing that the PI controllers provide good dynamic response and control performance.

Overall, the PI controllers ensure fast response, reduced steady-state error, and stable operation of the PMSM drive system.

3.2 Park Transformation

The Park transformation is a fundamental mathematical tool used in Field Oriented Control (FOC) to convert the stationary reference frame quantities into a rotating reference frame. This transformation enables effective control of the Permanent Magnet Synchronous Motor (PMSM) by simplifying the analysis of three-phase AC signals.

After applying the Clarke transformation, the three-phase stator currents (I_a, I_b, I_c) are converted into two-phase stationary components (I_α, I_β). The Park transformation further converts these stationary components into rotating reference frame components (I_d, I_q), which rotate synchronously with the rotor magnetic field.

The transformation is mathematically expressed as:

$$\begin{bmatrix} I_d \\ I_q \end{bmatrix} = \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} I_\alpha \\ I_\beta \end{bmatrix} \tag{3.5}$$

where θ is the rotor electrical angle obtained from the encoder.

The d-axis (I_d) represents the component of current aligned with the rotor flux, and it is responsible for controlling the magnetic flux in the motor. The q-axis (I_q) represents the component of current perpendicular to the rotor flux, and it is responsible for producing electromagnetic torque.

One of the key advantages of the Park transformation is that it converts time-varying sinusoidal quantities in the stationary frame into constant (DC) quantities in the rotating frame under steady-state conditions. This greatly simplifies the control process, as controlling DC quantities is easier compared to controlling AC signals.

In the context of FOC, this transformation allows the PMSM to be controlled in a manner similar to a separately excited DC motor, where flux and torque can be controlled independently. This decoupling of control improves system performance, reduces torque ripple, and enhances dynamic response.

3.2.1 Role in the Control System

In the implemented simulation model, the Park transformation is applied after the Clarke transformation block. The measured stator currents are first converted into α-β components and then transformed into d-q components using the rotor angle.

These d-q currents are then used as inputs to the PI controllers, which regulate the flux and torque independently. Accurate estimation of the rotor angle (θ) is critical for proper transformation, as any misalignment can result in incorrect current components and degraded system performance.

Thus, the Park transformation plays a crucial role in enabling efficient and precise control of the PMSM in electric bike applications.

3.3 Inverse Park Transformation

The inverse Park transformation is used to convert the voltage components from the rotating d-q reference frame back into the stationary α-β reference frame. This step is essential in the Field Oriented Control (FOC) process, as the inverter requires stationary frame voltage signals for proper operation.

After the PI controllers generate the control voltages (V_d and V_q) in the rotating reference frame, these voltages must be transformed back to the stationary frame before being applied to the inverter. This transformation ensures that the control signals are compatible with the physical three-phase system.

The inverse Park transformation is mathematically expressed as:

$$\begin{bmatrix} V_\alpha \\ V_\beta \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} V_d \\ V_q \end{bmatrix} \tag{3.6}$$

where θ is the rotor electrical angle obtained from the encoder.

In this transformation, the d-axis and q-axis voltage components are projected back onto the stationary α-β axes. Unlike the Park transformation, which aligns the reference frame with the rotor, the inverse Park transformation reverses this alignment, converting the rotating frame signals into stationary sinusoidal components.

These resulting α-β voltages are then further converted into three-phase voltages (V_a, V_b, V_c) using the inverse Clarke transformation. These three-phase voltage signals are used to generate Pulse Width Modulation (PWM) signals for controlling the inverter switches.

3.3.1 Role in the Control System

In the implemented simulation model, the inverse Park transformation is applied after the PI controllers. The voltage outputs (V_d and V_q) generated by the controllers are transformed into stationary frame voltages using the rotor angle.

This step is crucial because the inverter operates in the three-phase stationary reference frame and requires voltage inputs in this form. Without the inverse Park transformation, it would not be possible to apply the control signals to the motor.

Thus, the inverse Park transformation acts as a bridge between the control system (operating in the rotating frame) and the power system (operating in the stationary frame), ensuring proper implementation of the FOC strategy.

3.4 Clarke Transformation

The Clarke transformation is used to convert three-phase stator currents into a two-axis stationary reference frame. This transformation is an important step in the

Field Oriented Control (FOC) process, as it simplifies the analysis and control of the motor.

In a three-phase system, the stator currents (I_a, I_b, I_c) are sinusoidal and displaced by 120 degrees. These three variables are not independent, as their sum is zero in a balanced system. Therefore, they can be represented using only two orthogonal components in the stationary reference frame, known as the α-β frame.

The Clarke transformation is mathematically given by:

$$\begin{bmatrix} I_\alpha \\ I_\beta \end{bmatrix} = \frac{2}{3} \begin{bmatrix} 1 & -\frac{1}{2} & -\frac{1}{2} \\ 0 & \frac{\sqrt{3}}{2} & -\frac{\sqrt{3}}{2} \end{bmatrix} \begin{bmatrix} I_a \\ I_b \\ I_c \end{bmatrix} \tag{3.7}$$

The resulting components I_α and I_β represent the projection of the three-phase currents onto two perpendicular axes in the stationary reference frame.

The α axis is aligned with phase a, while the β axis is orthogonal to it. This transformation effectively reduces the complexity of the system by converting three-phase quantities into two-phase quantities without losing any information.

3.4.1 Physical Interpretation

The Clarke transformation can be understood as projecting the three-phase currents onto a two-dimensional coordinate system. Since the sum of the three-phase currents is zero in a balanced system, one of the components becomes redundant. Therefore, only two orthogonal components are sufficient to represent the system.

This transformation preserves the magnitude and power of the original three-phase system, ensuring that the energy content remains unchanged.

3.4.2 Role in the Control System

In the implemented simulation model, the Clarke transformation is applied to the measured stator currents (I_a, I_b, I_c) obtained from the motor. These currents are converted into α-β components, which are then used as inputs to the Park transformation.

This step is essential for reducing the complexity of the control system and preparing the signals for further transformation into the rotating reference frame.

Thus, the Clarke transformation serves as the first stage in the FOC algorithm, enabling efficient processing and control of the motor currents.

3.5 Inverse Clarke Transformation

The inverse Clarke transformation is used to convert the two-phase stationary reference frame quantities (α, β) back into three-phase quantities (a, b, c). This transformation is an essential step in the Field Oriented Control (FOC) process, as the inverter requires three-phase voltage inputs to drive the motor.

After the inverse Park transformation, the voltage components are obtained in the stationary α-β reference frame. These components must be converted into three-phase voltages before being applied to the inverter.

The inverse Clarke transformation is mathematically expressed as:

$$\begin{bmatrix} V_a \\ V_b \\ V_c \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ -\frac{1}{2} & \frac{\sqrt{3}}{2} \\ -\frac{1}{2} & -\frac{\sqrt{3}}{2} \end{bmatrix} \begin{bmatrix} V_\alpha \\ V_\beta \end{bmatrix} \tag{3.8}$$

The resulting voltages (V_a, V_b, V_c) represent the three-phase signals that are required to drive the inverter.

3.5.1 Physical Interpretation

The inverse Clarke transformation can be understood as reconstructing the original three-phase system from the two orthogonal components. The α and β components contain complete information about the system, and this transformation redistributes that information into three sinusoidal phase quantities.

These three-phase voltages are balanced and separated by 120 degrees, ensuring proper operation of the motor.

3.5.2 Role in the Control System

In the implemented simulation model, the inverse Clarke transformation is applied after the inverse Park transformation. The voltage components (V_α, V_β) obtained from the control system are converted into three-phase voltages (V_a, V_b, V_c).

These voltages are then used as reference signals for the PWM inverter, which generates switching pulses to control the motor.

Thus, the inverse Clarke transformation acts as the final step in converting control signals into a form suitable for practical implementation in the power stage of the system.

3.6 Inverter

The inverter is a key component in the electric bike drive system, responsible for converting the DC power from the battery into three-phase AC power required by the Permanent Magnet Synchronous Motor (PMSM). Since the battery provides a DC supply (72 V in this system), an inverter is essential to generate the alternating voltages needed for motor operation.

In this project, a three-phase Voltage Source Inverter (VSI) is used. The inverter consists of power electronic switches such as MOSFETs or IGBTs arranged in a bridge configuration. These switches are controlled using Pulse Width Modulation (PWM) techniques to produce the required AC output.

The control system generates three-phase reference voltages (V_a, V_b, V_c) through inverse Park and Clarke transformations. These reference signals are given as input to the PWM or Space Vector Pulse Width Modulation (SVPWM) block, which generates appropriate switching pulses for the inverter switches.

The inverter then uses these switching signals to convert the DC input into three-phase AC voltages with controlled magnitude and frequency. By varying the switching pattern, the inverter can regulate the motor speed and torque.

3.6.1 Working Principle

The inverter operates by rapidly switching the semiconductor devices ON and OFF in a controlled manner. The PWM technique adjusts the duty cycle of these switches to synthesize sinusoidal output voltages from the DC input.

In Space Vector PWM (SVPWM), the reference voltage vector is approximated using a combination of switching states, resulting in improved DC bus utilization and reduced harmonic distortion compared to conventional PWM methods.

The output of the inverter is a set of three-phase voltages that are applied to the stator of the PMSM, producing a rotating magnetic field that drives the motor.

3.6.2 Role in the Control System

In the implemented simulation model, the inverter acts as the interface between the control system and the motor. The control system operates in the d-q reference frame, while the inverter operates in the three-phase stationary frame.

The voltage commands obtained from the PI controllers are transformed into three-phase signals and fed into the SVPWM block. The inverter then generates the corresponding AC voltages required to drive the motor.

Thus, the inverter plays a crucial role in translating control signals into actual electrical power delivered to the motor.

3.6.3 Performance Considerations

The performance of the inverter directly affects the efficiency and quality of motor operation. Proper switching reduces harmonic distortion, minimizes losses, and improves overall system efficiency.

In the simulation results, the small ripples observed in torque and current waveforms are mainly due to the switching action of the inverter. However, the overall system remains stable, indicating effective inverter operation.

3.7 Hall Effect Sensor

The Hall Effect sensor is used to measure the rotor position and speed of the Permanent Magnet Synchronous Motor (PMSM). Accurate rotor position information is essential for implementing Field Oriented Control (FOC), as it is required for performing Park and inverse Park transformations.

The Hall Effect sensor operates based on the Hall Effect principle, which states that when a current-carrying conductor is placed in a magnetic field, a voltage is generated perpendicular to both the current and the magnetic field. This voltage is known as the Hall voltage and is used to detect the position of the rotor.

In the motor, multiple Hall sensors are placed at specific positions (typically 120° apart). As the rotor rotates, the magnetic field from the permanent magnets interacts with the sensors, producing digital signals that indicate the rotor position.

3.7.1 Working Principle

As the rotor rotates, the Hall sensors generate a sequence of digital signals corresponding to different rotor positions. These signals are decoded to determine the electrical angle (θ) of the rotor.

The rotor electrical angle is given by:

$$\theta_e = p \cdot \theta_m \tag{3.9}$$

where θ_e is the electrical angle, θ_m is the mechanical angle, and p is the number of pole pairs.

This electrical angle is used in the Park and inverse Park transformations to align the reference frame with the rotor magnetic field.

3.7.2 Role in the Control System

In the implemented simulation model, the Hall Effect sensor provides rotor position feedback required for coordinate transformations. The obtained angle is used to convert currents and voltages between stationary and rotating reference frames.

Accurate position sensing ensures proper decoupling of flux and torque components. Any error in position measurement can lead to incorrect transformation, resulting in reduced efficiency and poor motor performance.

3.7.3 Advantages

The Hall Effect sensor is widely used in electric vehicle applications due to its simplicity, reliability, and low cost. It provides sufficient accuracy for practical applications and enables effective implementation of FOC in PMSM drives.

Thus, the Hall Effect sensor plays a crucial role in ensuring accurate control and stable operation of the electric bike drive system.

3.8 Motor

The motor used in this project is a Permanent Magnet Synchronous Motor (PMSM) rated at 2 kW. PMSMs are widely used in electric vehicle applications due to their high efficiency, high power density, and excellent torque characteristics.

In a PMSM, the rotor contains permanent magnets that produce a constant magnetic field, while the stator carries three-phase windings supplied by the inverter. When three-phase AC currents flow through the stator windings, a rotating magnetic field is generated, which interacts with the rotor magnetic field and produces torque.

3.8.1 Mathematical Modelling

The PMSM is commonly modelled in the d-q reference frame for control purposes. The voltage equations in the d-q frame are given by:

$$V_d = R_s I_d + L_d \frac{dI_d}{dt} - \omega_e L_q I_q \tag{3.10}$$

$$V_q = R_s I_q + L_q \frac{dI_q}{dt} + \omega_e (L_d I_d + \lambda_m) \tag{3.11}$$

where:

- V_d, V_q are d-axis and q-axis voltages
- I_d, I_q are d-axis and q-axis currents
- R_s is stator resistance
- L_d, L_q are d-axis and q-axis inductances
- ω_e is electrical angular speed
- λ_m is flux linkage due to permanent magnets

The electromagnetic torque produced by the PMSM is given by:

$$T_e = \frac{3}{2} p \left[\lambda_m I_q + (L_d - L_q) I_d I_q \right] \tag{3.12}$$

where p is the number of pole pairs.

For surface-mounted PMSM, $L_d \approx L_q$, and the torque equation simplifies to:

$$T_e = \frac{3}{2} p \lambda_m I_q \tag{3.13}$$

This shows that the torque is directly proportional to the q-axis current (I_q).

3.8.2 Working Principle

The inverter supplies three-phase AC currents to the stator windings, creating a rotating magnetic field. The rotor magnets align with this rotating field, causing the rotor to rotate synchronously with the stator field.

In Field Oriented Control, the stator currents are transformed into d-q components, where the d-axis current controls flux and the q-axis current controls torque. This enables independent control of torque and flux, improving motor performance.

3.8.3 Role in the System

In the implemented simulation model, the PMSM acts as the main plant of the system. It receives three-phase voltages from the inverter and produces outputs such as stator currents, rotor speed, rotor position, and electromagnetic torque.

These outputs are fed back into the control system to ensure closed-loop operation. The motor performance is analyzed using parameters such as speed response, torque response, and rotor angle variation.

Thus, the PMSM plays a central role in the electric bike drive system, converting electrical energy into mechanical motion efficiently and reliably.

3.9 Simulation

The electric bike drive system is modelled and simulated in MATLAB/Simulink using a Field Oriented Control (FOC) strategy for a Permanent Magnet Synchronous Motor (PMSM). The simulation integrates different subsystems including the reference model, control system, inverter, motor, and feedback mechanisms.

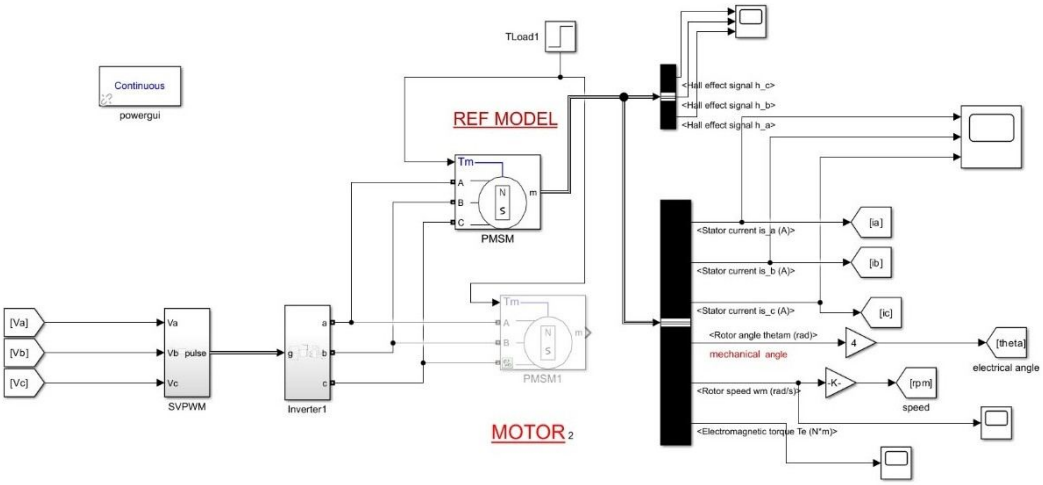


Figure 3.1: Matlab Simulink Model

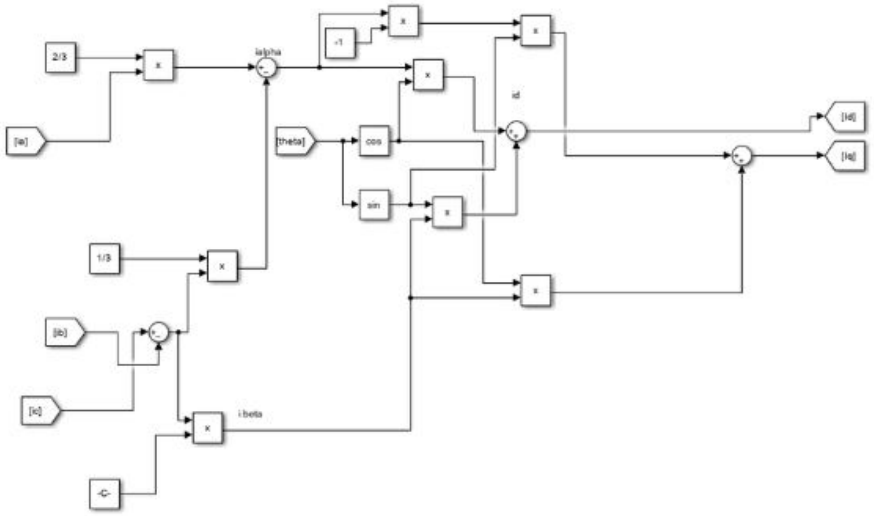


Figure 3.2: Clarke Park Transformation

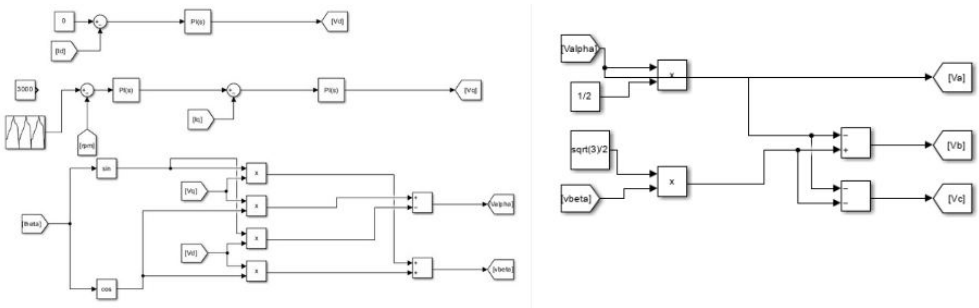


Figure 3.3: Inverse Clarke and Inverse Park Transformation

3.9.1 System Setup

The system is powered by a 72 V lithium-ion battery and uses a 2 kW PMSM as the main drive motor. The control strategy is implemented using FOC, which requires transformation blocks, PI controllers, and feedback signals.

The simulation model consists of the following major components:

- Reference model
- Clarke and Park transformation blocks
- PI controllers
- Inverse transformation blocks
- SVPWM-based inverter
- PMSM motor
- Hall Effect sensor for rotor position

3.9.2 Simulation Methology

The simulation begins with the reference model, which provides the input conditions such as desired speed or torque. A load torque is also applied to study the motor response under different operating conditions.

The control system generates three-phase voltage reference signals (V_a, V_b, V_c). These signals are processed by the Space Vector Pulse Width Modulation (SVPWM) block, which generates switching pulses for the inverter.

The inverter converts the DC input from the battery into three-phase AC voltages based on the switching signals. These voltages are applied to the PMSM, which converts electrical energy into mechanical output.

The motor produces outputs such as stator currents (I_a, I_b, I_c), rotor speed, rotor position, and electromagnetic torque.

The stator currents are measured and transformed into α-β components using Clarke transformation, and then into d-q components using Park transformation. These d-q currents are compared with reference values, and the resulting errors are processed by PI controllers.

The PI controllers generate control voltages (V_d, V_q), which are converted back into three-phase voltages using inverse Park and Clarke transformations. These voltages are again fed into the inverter, forming a closed-loop control system.

3.9.3 Feedback and Measurement

The rotor position is obtained using a Hall Effect sensor and converted into electrical angle for transformation purposes. The motor speed is measured and converted into RPM for analysis.

The electromagnetic torque and stator currents are also monitored to evaluate system performance. All these parameters are displayed using scope blocks for visualization.

3.9.4 Operating Conditions

The simulation is carried out under no-load and varying speed conditions. The system response is analyzed in terms of speed, torque, and rotor angle to evaluate the effectiveness of the control strategy.

The closed-loop operation ensures continuous correction of errors, resulting in accurate tracking of reference inputs and stable motor performance.

3.10 Result

The performance of the PMSM drive system using Field Oriented Control (FOC) was evaluated through simulation under different operating conditions. The results include analysis of speed, rotor angle, and electromagnetic torque under no-load and dynamic conditions.

3.10.1 No-Load Performance

The system was tested under no-load conditions to observe the dynamic response during startup.

3.10.2 Speed (Rpm) Tracking

The speed trace shows a rapid, smooth acceleration to the setpoint speed of 3000 Rpm.

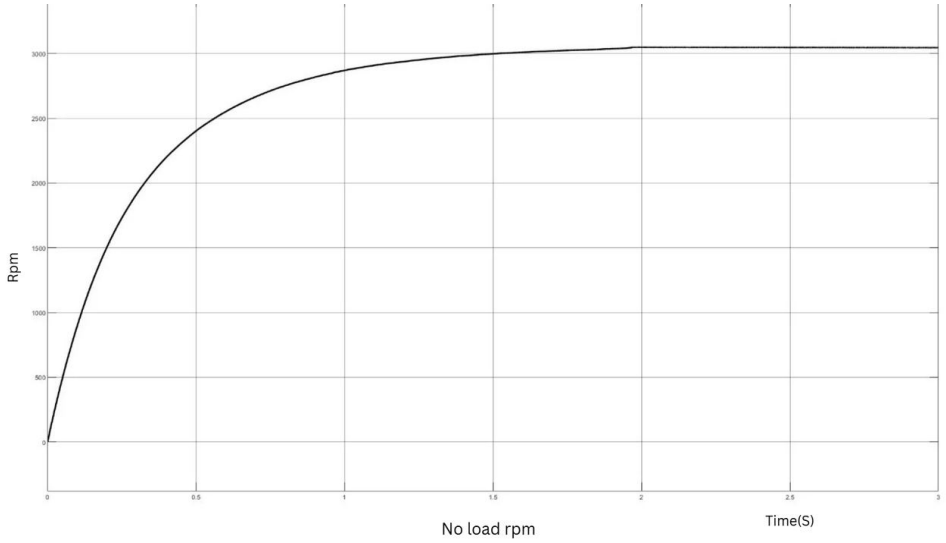


Figure 3.4: No-load Speed (Rpm) Response

The motor speed accurately tracks the reference, achieving a fast and smooth response with zero steady-state error, validating the effectiveness of the PI speed controller . The system reaches the 3000 RPM setpoint quickly without overshoot or oscillations, indicating proper decoupling of torque and flux through FOC. The close match between reference and actual speed demonstrates high tracking accuracy and robustness of the cascaded control structure. Overall, the no-load results confirm correct implementation of FOC, including PI tuning and Clarke–Park transformations, ensuring stable and precise speed control.

3.10.3 Electromagnetic Torque (T_e)

The torque graph illustrates the control's efficiency during startup.

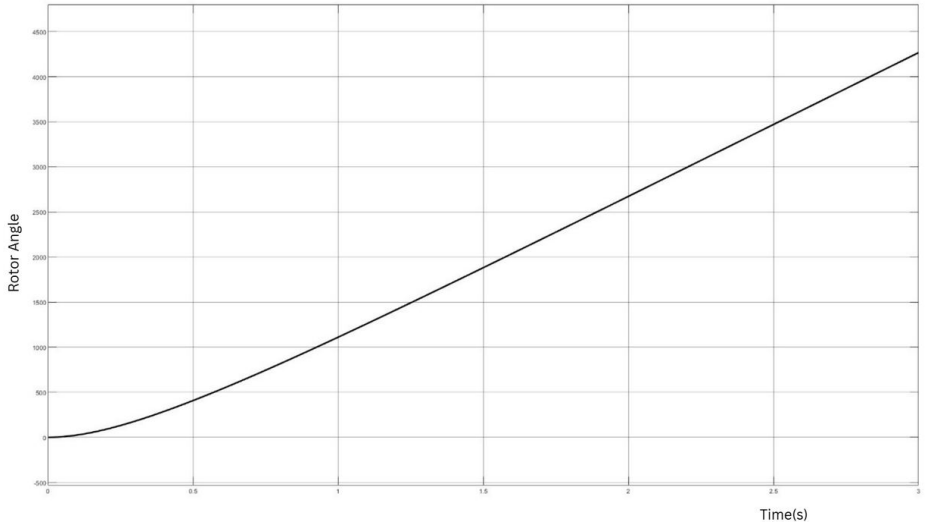


Figure 3.5: No-load Electromagnetic Torque (T_e) Response

The torque response shows a high initial peak during startup to overcome inertia, confirming strong acceleration capability of the FOC system. As the motor reaches the target speed, the torque quickly reduces to a minimal value, just enough to compensate for friction and losses. This demonstrates efficient coordination between speed and current control loops. Overall, the system provides only the required torque at any instant, validating high efficiency, precise control, and effective torque–flux decoupling.

3.10.4 Detailed Analysis of No-Load Stator Currents

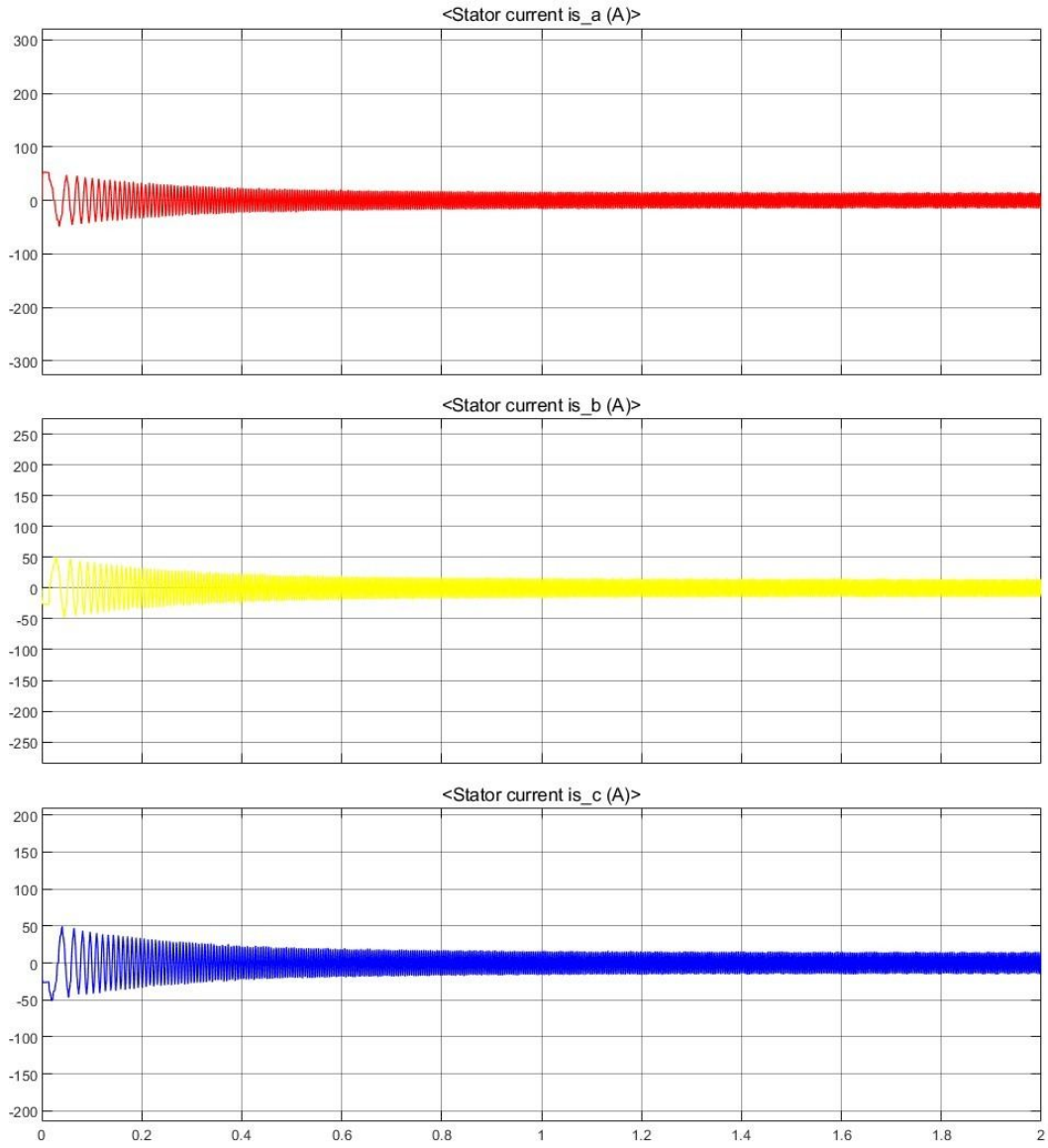


Figure 3.6: Stator currents under no-load condition.

The no-load stator current waveform shows the motor's dynamic behavior during startup and stabilization over 2 seconds. During the transient phase (0–0.5 s), a high initial current surge appears in all three phases to generate starting torque and overcome inertia, followed by a rapid damped decay, indicating effective current regulation and suppression of overshoot. In steady-state (0.5–2 s), the currents settle to a low amplitude (below ± 50 A), as only magnetizing current and internal losses are supplied. The waveforms remain balanced, symmetrical, and 120° phase-shifted, confirming smooth and stable operation under no-load conditions.

3.10.5 Detailed Analysis of On-Load Stator Currents

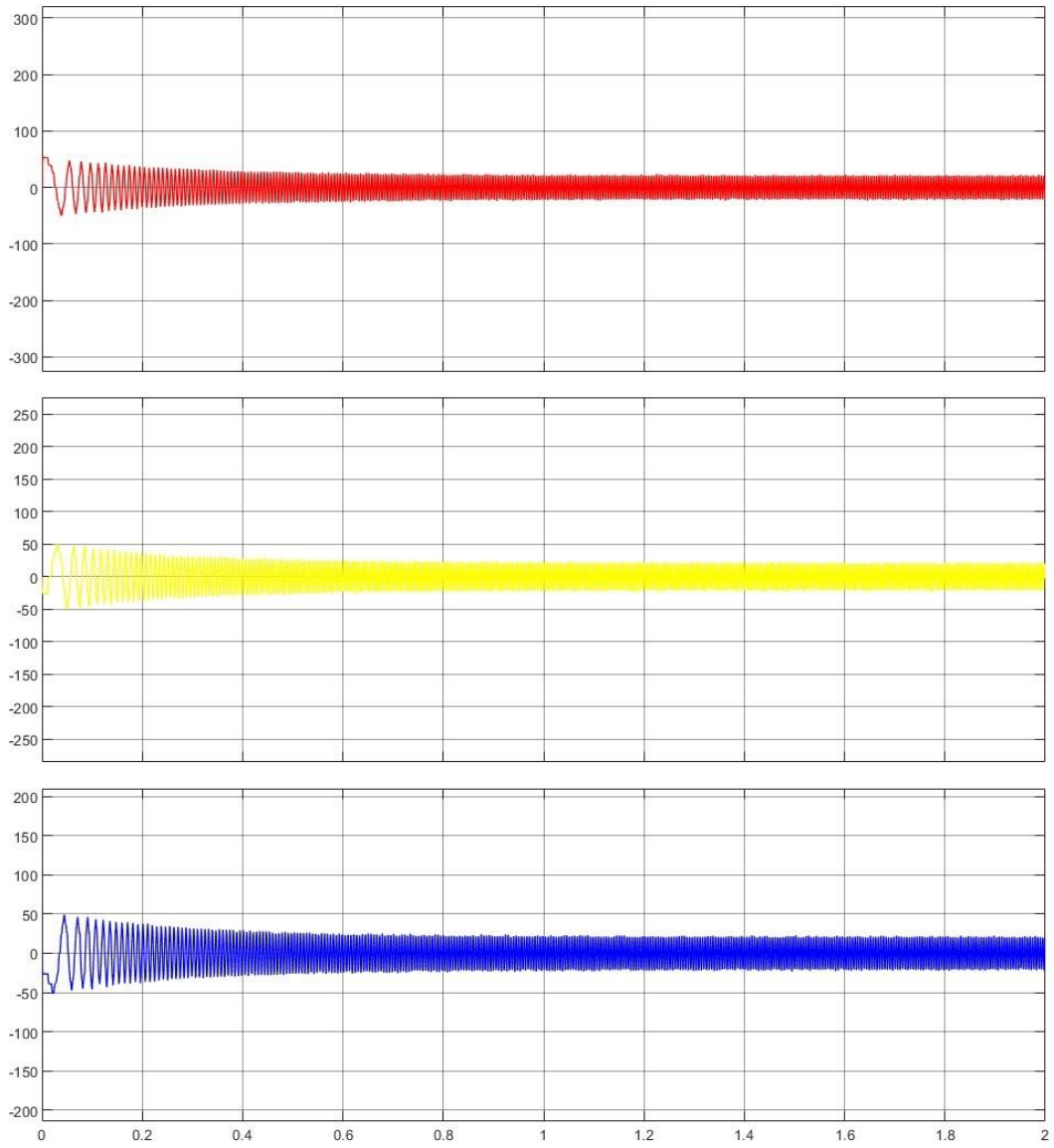


Figure 3.7: Stator currents under operating load condition.

The on-load stator current waveform shows the motor's behavior during startup and stabilization over 2 seconds. During the transient phase (0–0.5 s), a high initial current surge is observed in all phases to overcome combined inertia and mechanical load, followed by a rapid damped decay, indicating efficient current control and quick stabilization. In steady-state (0.5–2 s), the current amplitude remains higher (around ±50 A) compared to no-load, as continuous power is required to produce torque against the load. The steady current level reflects the torque demand, while the balanced, symmetrical waveforms with 120° phase shift confirm smooth and stable operation under load.

3.10.6 Performance Under Repeating Sequence Command

Speed (Rpm) Tracking

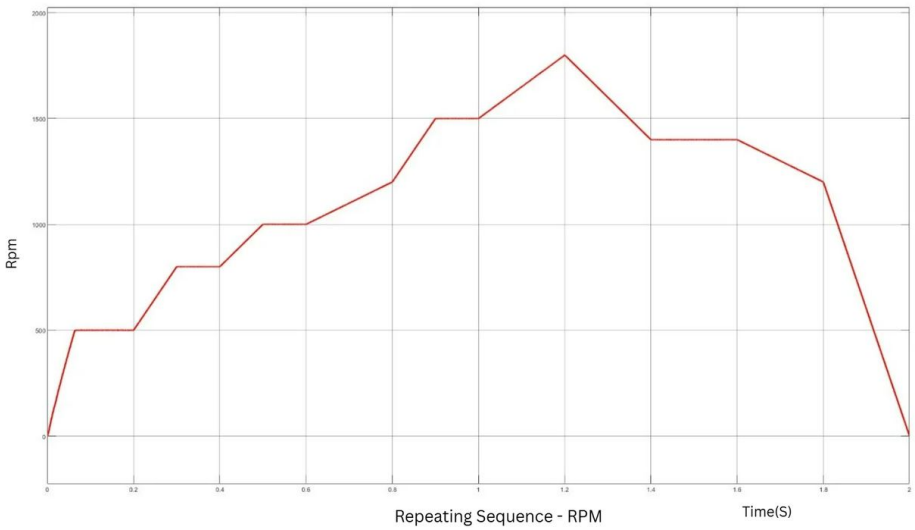


Figure 3.8: Repeating Sequence - Rpm Tracking

The motor speed closely matches the reference across all dynamic conditions, including rapid acceleration, deceleration, constant-speed regions, and step changes, with negligible delay. The near-perfect overlap confirms high bandwidth, fast response, and accurate tracking. This validates the strength of FOC, where real-time torque–flux decoupling is maintained through Clarke and Park transformations, ensuring independent control of Id and Iq. The PI controllers exhibit high agility by instantly adjusting voltage commands without overshoot or error. Overall, the system demonstrates precise, stable, and high-performance speed control suitable for dynamic applications like EVs and robotics.

Electromagnetic Torque (T_e)

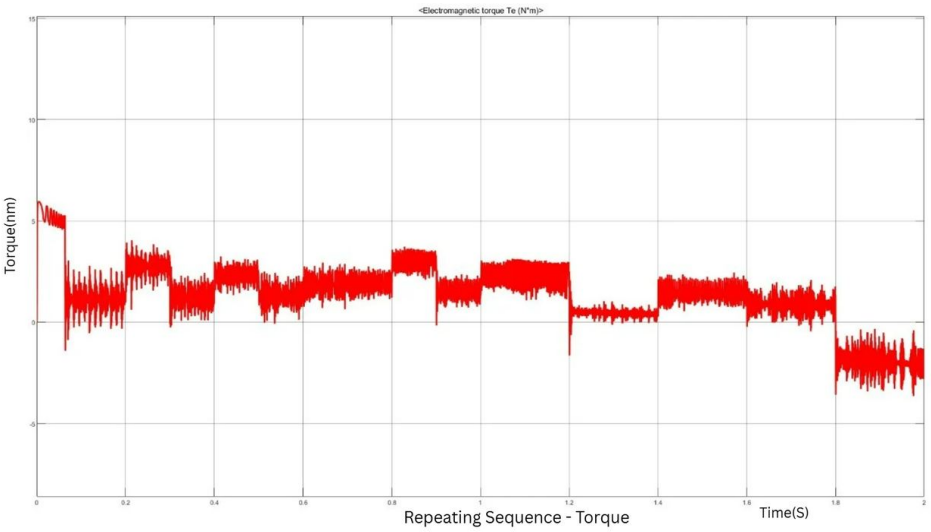


Figure 3.9: Repeating Sequence - Torque Response

The torque response closely follows speed commands, providing appropriate positive torque for acceleration and negative torque for deceleration. During acceleration, torque rises due to increased Iq, enabling rapid response to overcome inertia. During deceleration, torque becomes negative (regenerative braking) with instant reversal of Iq, ensuring precise speed tracking. At constant speed, torque quickly drops to near zero, supplying only the minimum required to overcome losses, demonstrating high efficiency. Overall, FOC ensures that speed errors are instantly converted into proportional torque, with perfect torque–flux decoupling enabling fast, accurate, and efficient tracking.

Detailed Analysis of Electromagnetic Torque After Inertia Change

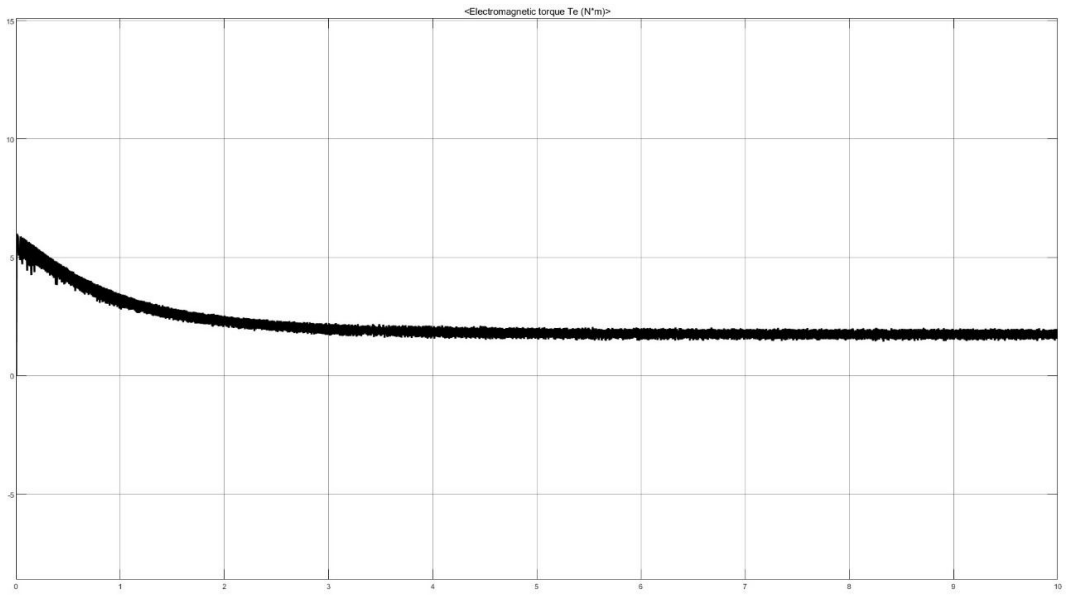


Figure 3.10: Electromagnetic Torque response after changing the motor's inertia value.

The torque response over 10 seconds shows the effect of increased inertia. During the transient phase (0–3 s), a high initial torque of about 7.5 N·m is generated to accelerate the load. The torque then decays exponentially over 3 seconds, with the longer decay indicating higher inertia, while smooth reduction confirms proper controller action without overshoot. In steady-state (3–10 s), the torque stabilizes at a low value of around 2.5 N·m, which balances mechanical load, friction, and windage losses. The minimal ripple in torque confirms stable operation and effective control, ensuring smooth power delivery despite increased inertia.

Detailed Analysis of Motor Speed (RPM) After Inertia Change

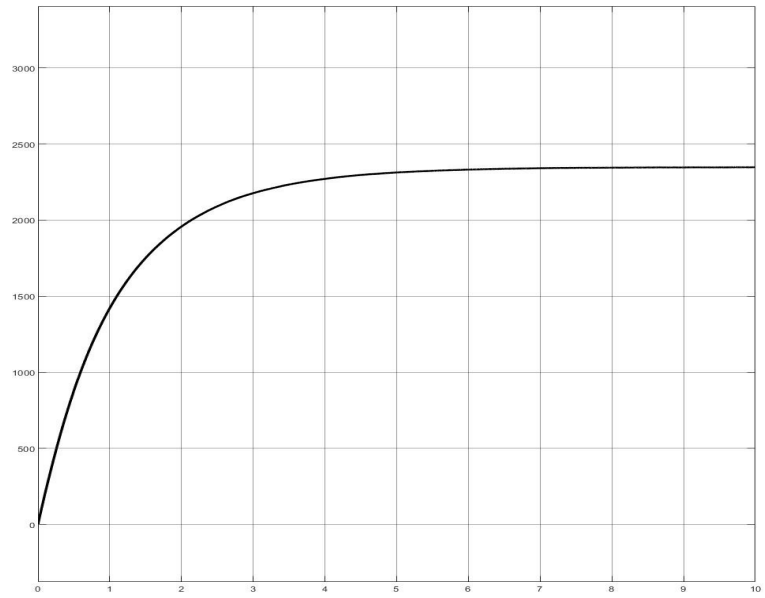


Figure 3.11: Motor speed (RPM) response after changing the inertia value.

The motor speed response over 10 seconds shows the effect of increased inertia. During the acceleration phase (0–5 s), the speed rises smoothly from 0 RPM, but with a slower rate due to higher inertia, increasing the time required to reach the setpoint. The response shows no overshoot, indicating a well-tuned speed control loop. In steady-state (5–10 s), the speed stabilizes at approximately 2350 RPM with a perfectly flat curve, confirming zero steady-state error. This demonstrates accurate, stable, and robust speed control even under increased inertia conditions.

3.10.7 Detailed Analysis of Stator Currents After Increasing Inertia



Figure 3.12: Stator currents after increasing the mechanical inertia.

The stator current response over 10 seconds shows the effect of increased inertia. During the transient phase (0–2 s), a large initial current surge occurs in all phases to generate the high starting torque required for accelerating the heavy inertia load. The decay of the current is slightly prolonged (up to 2 s), indicating sustained effort due to higher inertia, while the smooth damping confirms stable and effective control. In steady-state (2–10 s), the currents settle into balanced sinusoidal waveforms with relatively low amplitude (around ± 50 A), demonstrating stable and smooth operation under a balanced three-phase supply despite increased inertia.

3.11 Summary

This chapter presented the modelling, simulation, and analysis of a Permanent Magnet Synchronous Motor (PMSM) drive system using Field Oriented Control (FOC) for an electric bike application.

The system was modelled using transformation techniques including Clarke and Park transformations, along with their inverse counterparts, to convert three-phase quantities into a rotating reference frame and back. This enabled decoupled control of flux and torque components.

PI controllers were implemented to regulate the d-axis and q-axis currents, ensuring accurate control of motor flux and torque. The inverter, controlled using Space

Vector Pulse Width Modulation (SVPWM), converted the DC supply into controlled three-phase AC voltages required by the motor. A Hall Effect sensor was used to obtain rotor position feedback, which is essential for coordinate transformations and proper synchronization.

The simulation was carried out in MATLAB/Simulink using system parameters corresponding to a 72 V battery and a 2 kW PMSM. The results obtained from the simulation demonstrated smooth speed response, stable torque characteristics, and accurate rotor position tracking.

Under no-load conditions, the motor showed fast acceleration and stable steady-state operation. During dynamic conditions, the system successfully tracked varying speed references with minimal delay. The torque response was consistent with speed variations, showing proper control during both acceleration and deceleration.

Overall, the simulation results confirm that the Field Oriented Control strategy provides efficient, stable, and reliable performance for PMSM drives. The system achieves effective decoupling of torque and flux, fast dynamic response, and accurate control, making it well-suited for electric bike applications.

CHAPTER 4

SYSTEM DESIGN AND ARCHITECTURE

The electrical architecture of the retrofitted electric motorcycle integrates two distinct subsystems: the High Voltage (HV) traction system and the Low Voltage (LV) control and auxiliary system. Together, these subsystems form a complete power delivery and management platform that enables safe, efficient, and reliable operation. The design is centred on a **QS Motor 2000 W PMSM** driven by a **FarDriver ND72340** motor controller, powered by a **72 V / 30 Ah lithium-ion battery pack**.

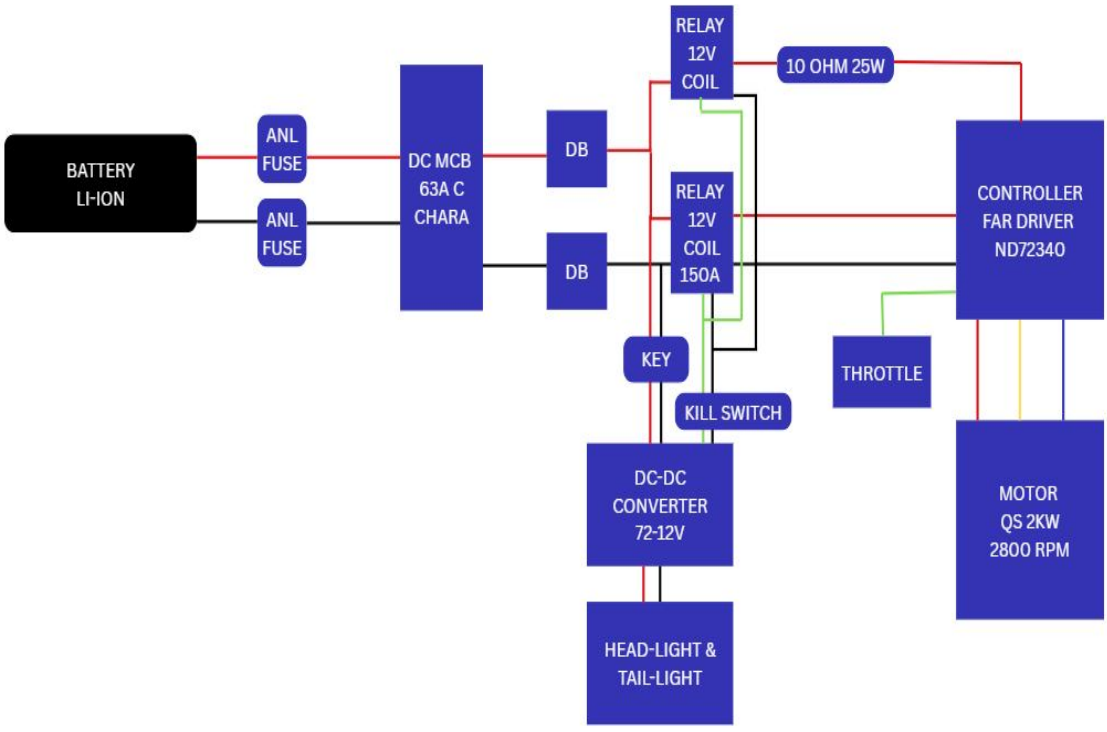


Figure 4.1: HV and LV Architecture

4.1 High Voltage (HV) System

The HV system operates at a nominal voltage of **72 V DC** and constitutes the primary traction power path. It is responsible for delivering electrical energy from the battery pack to the traction motor through a series of protection, switching, and conversion elements.

4.1.1 Battery



Figure 4.2: Battery Cell

Cell Specification

The battery pack uses lithium-ion cells rated at 5000 mAh (5 Ah), 3.6 V nominal, with a maximum continuous discharge rate of 3C.

Pack Configuration

The cells are arranged in a **20S6P** configuration (20 cells in series, 6 parallel strings), yielding:

$$V_{\text{nominal}} = 20 \times 3.6 = 72\,\text{V} \tag{4.1}$$

$$V_{\text{max}} = 20 \times 4.2 = 84\,\text{V} \tag{4.2}$$

$$V_{\text{min}} = 20 \times 2.8 = 56\,\text{V} \tag{4.3}$$

$$C_{\text{pack}} = 6 \times 5 = 30\,\text{Ah} \tag{4.4}$$

$$E_{\text{pack}} = V_{\text{nominal}} \times C_{\text{pack}} = 72 \times 30 = 2160\,\text{Wh} = 2.16\,\text{kWh} \tag{4.5}$$

Maximum Discharge Current

Each cell supports a continuous discharge of 3C:

$$I_{\text{cell,max}} = 3\text{C} \times 5\,\text{Ah} = 15\,\text{A} \tag{4.6}$$

$$I_{\text{pack,max}} = 6 \times 15 = 90\,\text{A} \tag{4.7}$$

This exceeds the maximum controller demand of 80 A, ensuring no cell-level stress under peak conditions.

Connector and Interface

The battery uses an **Anderson SB50** connector rated for 50 A continuous service. This connector provides a secure, polarity-keyed, and vibration-resistant interface between the battery pack and the HV bus.

4.1.2 Battery Management System (BMS)



Figure 4.3: Battery Management System

The **DALY Li-ion BMS 24S 100A** is used to protect and manage each cell in the pack. Key functions include:

- Individual cell voltage monitoring and over/under-voltage protection
- Over-temperature and under-temperature protection via dual NTC sensors (NTC-A, NTC-B)
- Overcurrent and short-circuit protection on charge and discharge paths
- Active cell balancing at up to 1 A to equalise cell voltages
- Communication interfaces: CAN/RS-485, UART, KEY, DIO

4.1.3 ANL Fuse



Figure 4.4: ANL Fuse

An **ANL fuse rated at 80 A** is installed as the primary short-circuit protection device at the battery output. The fuse is selected based on the total system maximum current and the cable current-carrying capacity (12 AWG silicone wire, rated approximately 40–60 A depending on ambient conditions; dual-conductor routing effectively doubles capacity).

If the instantaneous current exceeds 80 A due to a fault, the fusible element melts, permanently disconnecting the circuit and protecting downstream components.

The maximum fault current the fuse must interrupt is bounded by:

$$I_{\text{fault,max}} = \frac{V_{\text{pack,max}}}{R_{\text{wiring}}} \tag{4.8}$$

where R_{wiring} includes cable and connector resistance (typically a few mΩ for short high-current cables), which results in very large prospective fault currents. The ANL fuse provides the required interrupting capacity for such conditions.

4.1.4 DC Miniature Circuit Breaker (MCB)



Figure 4.5: DC Miniature Circuit Breaker

A **DC Double-Pole MCB, 63 A, Type-C characteristic curve (SB202M-C63)** is installed downstream of the ANL fuse. It provides:

- Manual ON/OFF switching for the HV bus
- Isolation of the system during maintenance
- Overload protection with Type-C trip characteristic (suitable for motor loads with moderate inrush)

The MCB rating of 63 A is selected to be greater than the continuous operational current (≈30 A rated) but within the cable capacity

4.1.5 Distribution Bus (DB)

The Distribution Bus acts as a HV junction point downstream of the MCB. It splits the 72 V supply into two branches:

1. **Branch 1 — LV Supply:** Routes power to the DC-DC buck converter for the 12 V low-voltage system.

2. **Branch 2 — Motor Drive:** Routes power through the pre-charge circuit and main relay to the motor controller.

4.1.6 Pre-Charge Circuit

The FarDriver ND72340 motor controller contains large internal filter capacitors that act as a near-short-circuit load at the instant the HV supply is first connected. Without current limiting, this inrush causes:

- Electrical arcing and welding at relay contacts
- High di/dt stress on wiring and connectors
- Potential damage to controller input stages

A **10 Ω, 25 W wire-wound resistor** is placed in series with the HV supply during startup to limit inrush current:

$$I_{\text{inrush}} = \frac{V_{\text{bus}}}{R_{\text{precharge}}} = \frac{72}{10} = 7.2\,\text{A} \tag{4.9}$$

The power dissipated in the resistor during pre-charge (transient event):

$$P_{\text{precharge}} = I_{\text{inrush}}^2 \times R = 7.2^2 \times 10 = 518.4\,\text{W} \quad \text{(instantaneous, transient only)} \tag{4.10}$$

The 25 W continuous rating of the resistor is adequate since the pre-charge duration is typically less than 1–2 seconds. Once the capacitors reach approximately 95% of bus voltage, the main relay is energised, bypassing the pre-charge resistor.

The RC time constant for the pre-charge circuit is:

$$\tau = R_{\text{precharge}} \times C_{\text{controller}} \tag{4.11}$$

where $C_{\text{controller}}$ is the equivalent capacitance of the controller input stage (typically 2000–6800 μF for controllers in this class). For a representative value of 4700 μF:

$$\tau = 10 \times 4700 \times 10^{-6} = 0.047\,\text{s} \approx 47\,\text{ms} \tag{4.12}$$

Full pre-charge to 95% occurs at $t \approx 3\tau \approx 141\,\text{ms}$.

4.1.7 Main Relay



Figure 4.6: Relay

A **high-current relay rated at 150 A, with a 12 V coil** is used as the main contactor. Once pre-charge is complete, the LV control circuit energises the relay coil, closing the main contacts and connecting the full HV supply to the motor controller.

Key specifications:

- Contact rating: 150 A DC
- Coil voltage: 12 V DC (driven from LV system)
- Normally open (NO) configuration

The relay is de-energised by the kill switch in the event of an emergency, immediately disconnecting HV power from the controller.

4.1.8 Motor Controller — FarDriver ND72340



Figure 4.7: FarDriver ND72340

The **FarDriver ND72340** is a high-performance sinusoidal FOC (Field-Oriented Control) motor controller designed for PMSM and BLDC motors in electric vehicle applications. It acts as the power inverter, converting 72 V DC from the battery into a controlled three-phase AC output for the motor.

4.1.9 Motor



Figure 4.8: 2KW PMSM Motor

The motor used in this project is a Permanent Magnet Synchronous Motor (PMSM) rated at 2 kW. PMSMs are widely used in electric vehicle applications due to their high efficiency, high power density, and superior torque characteristics.

Construction

The PMSM consists of two main parts: the stator and the rotor.

- **Stator:** The stator contains three-phase windings. When supplied with AC current, it produces a rotating magnetic field.
- **Rotor:** The rotor contains permanent magnets that generate a constant magnetic field. This eliminates the need for external excitation.

Working Principle

When three-phase AC currents are supplied to the stator, a rotating magnetic field is produced. The rotor magnets align with this field and rotate at the same speed, resulting in synchronous operation.

The synchronous speed is given by:

$$N_s = \frac{120f}{P} \tag{4.13}$$

where f is the supply frequency and P is the number of poles.

Mathematical Model

The PMSM is modelled in the d-q reference frame for control purposes. The voltage equations are:

$$V_d = R_s I_d + L_d \frac{dI_d}{dt} - \omega_e L_q I_q \tag{4.14}$$

$$V_q = R_s I_q + L_q \frac{dI_q}{dt} + \omega_e (L_d I_d + \lambda_m) \tag{4.15}$$

The electromagnetic torque is given by:

$$T_e = \frac{3}{2} p \lambda_m I_q \tag{4.16}$$

This shows that the torque is directly proportional to the q-axis current (I_q).

Control using FOC

In Field Oriented Control, the stator currents are transformed into d-q components, where the d-axis current (I_d) controls the flux and the q-axis current (I_q) controls the torque. For efficient operation, I_d is maintained close to zero, while I_q is varied to regulate the electromagnetic torque of the motor.

Application in Electric Bike

In this system, the PMSM acts as the main drive motor. It receives three-phase AC power from the controller and converts electrical energy into mechanical energy. The output is transmitted to the wheel through a chain drive.

The motor operates under varying conditions such as acceleration, steady-state operation, and deceleration. The FOC strategy ensures smooth speed control, stable torque response, and efficient performance.

4.1.10 Summary of HV System

The HV system ensures efficient power delivery from the battery to the motor while incorporating necessary protection and control mechanisms. The integration of fuse, MCB, pre-charge circuit, and relay ensures safe operation under all conditions.

4.2 Low Voltage (LV) System

The LV system operates at **12 V DC** and manages all control, signalling, and auxiliary functions of the motorcycle. It interfaces between the rider's inputs and the HV power system to ensure controlled, safe operation.

4.2.1 DC-DC Buck Converter

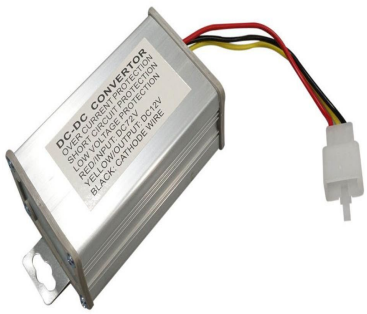


Figure 4.9: DC-DC Buck Converter

A **72 V to 12 V DC-DC buck converter** steps down the HV bus voltage to a stable 12 V supply for all LV loads. The converter also provides a secondary regulated **5 V**

output for microcontroller (Arduino) supply.

$$\eta_{\text{buck}} = \frac{P_{\text{out}}}{P_{\text{in}}} \times 100; \quad I_{12\text{V}} = \frac{P_{\text{LV loads}}}{12} \tag{4.17}$$

The converter is rated at 150 W and supports the combined auxiliary load including relay coils, Arduino, indicator lights, headlight, and taillight.

4.2.2 Arduino Microcontroller

An **Arduino** microcontroller serves as the central logic unit of the LV system. It receives a 5 V regulated supply from the DC-DC converter and manages:

- Three-speed RPM control logic (Eco / Normal / Sport)
- Signal processing for throttle input
- Relay energisation sequencing (pre-charge → main relay logic)
- Kill switch interrupt handling
- Indicator and auxiliary load switching via L298N motor driver outputs

4.2.3 L298N Motor Driver

The **L298N dual H-bridge motor driver** is used in the LV system not for traction motor driving, but for switching auxiliary loads under Arduino control. It interfaces between the 5 V Arduino logic outputs and 12 V relay coil / auxiliary load circuits, providing current amplification and back-EMF protection for inductive loads.

4.2.4 Control Inputs

- **Throttle:** Hall-effect based, 0–5 V output. Signal conditioned to the controller's required 1.1–4.05 V operating range. Provides proportional torque demand.
- **Key Switch:** Main system activation switch. When closed, it enables the DC-DC converter output and powers the Arduino and relay control circuitry.
- **Kill Switch:** Safety interrupt that immediately de-energises both pre-charge and main relay coils, disconnecting HV power from the controller and stopping the motor.

- **3-Speed Mode Selector:** Selects between Eco, Normal, and Sport RPM limits via Arduino logic, which sends the corresponding mode signal to the FarDriver controller.

4.2.5 Relay Control and Switching Logic

The startup sequence managed by the Arduino and LV relay logic is:

1. Key switch ON → DC-DC converter activates → Arduino boots.
2. Arduino energises **pre-charge relay** (12 V coil), current flows through 10 Ω resistor to charge controller capacitors.
3. After $\approx 3\tau \approx 141$ ms, Arduino energises **main relay**, completing the full HV circuit.
4. Throttle signal is now active; motor controller begins normal FOC operation.
5. Kill switch activation → Arduino immediately de-energises both relays.

4.2.6 Safety and Control Functions



Figure 4.10: Kill Switch

The LV system ensures safe operation by controlling system activation and shutdown. The key switch prevents unauthorized use, while the kill switch provides immediate shutdown capability. The relay control logic ensures that high-voltage power is applied only after safe conditions are met.

4.3 Cable and Connector Selection

4.3.1 High Voltage Wiring

- **Cable type:** 12 AWG silicone-insulated stranded copper wire
- **Rated current:** ≈40–60 A per conductor (derated for bundled routing)
- **Voltage rating:** 600 V (suitable for 84 V max bus)
- **Temperature rating:** 200 °C silicone insulation
- **Flexibility:** High-flex silicone jacket for vibration resistance

4.4 System Parameters

The key electrical parameters of the system are listed in Table 4.1.

Table 4.1: System Parameters

Parameter	**Value**
Battery Type	Lithium-ion
Nominal Voltage	72 V
Battery Capacity	30 Ah
Energy Capacity	2.16 kWh
Motor Type	PMSM
Motor Power	2 kW
Controller	FarDriver ND72340
Rated Current	30 A
Maximum Current	60 A
DC-DC Converter	72V to 12V, 150 W

4.5 Summary

This chapter presented the complete system design and architecture of the electric motorcycle's electrical system. The HV system, operating at 72 V DC, delivers traction power from the 20S6P lithium-ion battery pack through a protective chain comprising an 80 A ANL fuse, a 63 A Type-C DC MCB, a pre-charge circuit with a 10 Ω/25 W resistor, and a 150 A main relay before reaching the FarDriver ND72340 FOC motor controller and the 2 kW QS Motor PMSM.

The LV system at 12 V, powered by a DC-DC buck converter, manages all control and auxiliary functions through an Arduino microcontroller and L298N driver, providing three-speed RPM control, relay switching logic, kill-switch safety, and auxiliary lighting and signalling loads. The interaction of HV and LV systems ensures

that full traction power is only applied after safe pre-charge conditions are confirmed, protecting relay contacts, wiring, and the controller from destructive inrush events. Together, the complete architecture delivers safe, efficient, and controllable electric propulsion within the competition rulebook constraints.

CHAPTER 5

MOTOR TUNING

5.1 Introduction

Motor tuning refers to the process of configuring the motor controller parameters to achieve the desired performance characteristics of the electric motorcycle. For the FarDriver ND72340 operating the QS Motor 2000 W PMSM, tuning involves setting throttle response curves, current limits, speed modes, and phase current ratios to balance acceleration performance, thermal management, and rideability.

Proper tuning is critical because the same motor and controller hardware can exhibit significantly different performance depending on the configured parameters. This chapter details all tuning parameters applied to the system and provides a comparison between the motor's rated and operational specifications.

5.1.1 Gear Ratio Impact on Torque

The system uses a **14T front sprocket and 42T rear sprocket**, giving:

$$G_{\text{system}} = \frac{42}{14} = 3:1 \tag{5.1}$$

The lower gear ratio trades maximum torque for higher top speed. Wheel speed and vehicle speed at rated motor speed:

$$N_{\text{wheel}} = \frac{N_{\text{motor}}}{G} = \frac{2800}{3} = 933\,\text{RPM} \tag{5.2}$$

$$v = \frac{2\pi r N_{\text{wheel}}}{60} = \frac{2\pi \times 0.307 \times 933}{60} \approx 30\,\text{m/s} = 108\,\text{km/h} \tag{5.3}$$

Note: actual top speed is limited by controller speed limits set in the three-speed mode profiles (Eco / Normal / Sport), with Sport mode targeting approximately 70–80 km/h as per the project objective.

5.2 FarDriver ND72340 Tuning Parameters

5.2.1 Throttle Configuration

Table 5.1: Throttle Tuning Parameters

Parameter	**Value**	**Description**
Throttle Response Mode	Sport (Mode 1)	Aggressive acceleration curve
Throttle Acc Step	12	Rate of torque increase on acceleration
Throttle Dec Step	20	Rate of torque reduction on deceleration
Throttle Low Voltage	1.1 V	Zero-torque threshold
Throttle High Voltage	4.05 V	Full-torque threshold

Throttle Response Analysis

The throttle input voltage is mapped to a torque demand signal. The effective control range is:

$$\Delta V_{\text{throttle}} = V_{\text{high}} - V_{\text{low}} = 4.05 - 1.1 = 2.95\,\text{V} \tag{5.4}$$

The torque command is linearly interpolated across this range:

$$T_{\text{demand}}(\%) = \frac{V_{\text{throttle}} - V_{\text{low}}}{V_{\text{high}} - V_{\text{low}}} \times 100 \tag{5.5}$$

A higher **Throttle Acc Step** means the controller ramps up current more slowly per control cycle, providing smooth yet responsive acceleration. The **Throttle Dec Step** allows for a faster regenerative/engine-braking effect on throttle release.

5.2.2 Current Limits

Table 5.2: Current Limit Configuration

Parameter	Value	Remarks
Maximum Line Current	75 A	Continuous HV bus limit
Maximum Phase Current	320 A	Peak phase (inverter output)
Boost Line Current	80 A	Short-duration boost
Boost Phase Current	340 A	Short-duration phase boost

The distinction between line current and phase current is important:

- **Line current (I_{line}):** DC bus current drawn from the battery. Directly governs battery discharge rate and is limited to protect the BMS and fuse.
- **Phase current (I_{phase}):** AC current in each motor winding, which can be significantly higher than line current at low duty cycles. High phase current at low speed produces maximum torque during starting.

The relationship between line and phase current at a given modulation index m is approximately:

$$I_{\text{line}} \approx \frac{3}{2} \cdot m \cdot I_{\text{phase}} \cdot \cos\phi \tag{5.6}$$

where ϕ is the power factor angle. At low speeds (low m), phase current can greatly exceed line current, enabling high starting torque without high battery discharge.

5.2.3 Low Speed Phase Ratio

$$\text{Low Speed Phase Ratio} = 140\% \tag{5.7}$$

This parameter instructs the controller to allow phase current to reach 140% of the configured maximum phase current threshold specifically during low-speed operation (typically below a configurable speed threshold). This provides enhanced starting torque and hill-climb capability:

$$I_{\text{phase,low-speed}} = 1.40 \times 320 = 448\,\text{A} \quad \text{(brief transient burst)} \tag{5.8}$$

$$T_{\text{peak,low-speed}} \propto I_{\text{phase,low-speed}} \implies \text{enhanced launch torque} \tag{5.9}$$

5.2.4 Three-Speed Mode Profiles

The FarDriver ND72340 and Arduino-based speed selector implement three operating modes:

Table 5.3: Three-Speed Mode RPM Profiles

Mode	**Target RPM**	**Est. Speed (km/h)**	**Use Case**
Eco	≈500 RPM	≈18–20	City slow, battery conservation
Normal	≈700 RPM	≈25–28	General urban riding
Sport	≈1000 RPM+	≈35–40	Performance / highway

Note: These RPM values reflect motor shaft speed at the three selectable controller speed limits. Actual wheel speed depends on the 1:3 gear ratio and tyre diameter. Full-throttle Sport mode can exceed these soft limits up to the programmed maximum.

5.3 Thermal and Protection Analysis

5.3.1 Controller Thermal Considerations

The FarDriver ND72340 dissipates heat primarily in its MOSFET power stage. At rated operation:

$$P_{\text{loss,controller}} = P_{\text{in}} - P_{\text{out}} = \frac{P_{\text{rated}}}{\eta} - P_{\text{rated}} = \frac{2000}{0.88} - 2000 = 272.7\,\text{W} \tag{5.10}$$

At boost conditions (80 A line current):

$$P_{\text{loss,boost}} = V \times I_{\text{boost}} \times (1 - \eta) = 72 \times 80 \times 0.12 = 691.2\,\text{W} \tag{5.11}$$

Adequate heatsinking and airflow must be maintained to prevent triggering thermal protection.

5.3.2 Motor Thermal Considerations

Motor copper losses at rated current:

$$P_{\text{Cu}} = 3 \times I_{\text{rated}}^2 \times R_s \tag{5.12}$$

where R_s is stator winding resistance per phase. Air cooling by natural convection is relied upon; the motor housing dissipates heat through its finned outer surface. The 88% efficiency at rated load gives a total motor loss of:

$$P_{\text{loss,motor}} = P_{\text{rated}} \times \left(\frac{1}{\eta} - 1\right) = 2000 \times \left(\frac{1}{0.88} - 1\right) = 272.7\,\text{W} \tag{5.13}$$

5.4 Cable Sizing Verification

The 12 AWG silicone wire used for HV connections has a DC resistance of approximately 5.2 mΩ/m. For a cable run of approximately 1.5 m at maximum current of 80 A:

$$V_{\text{drop}} = I \times R_{\text{cable}} = 80 \times (5.2 \times 10^{-3} \times 1.5) = 80 \times 7.8 \times 10^{-3} = 0.624\,\text{V} \tag{5.14}$$

$$\text{Voltage drop } \% = \frac{0.624}{72} \times 100 = 0.87\% \quad \text{(acceptable)} \tag{5.15}$$

$$P_{\text{cable,loss}} = I^2 \times R = 80^2 \times 7.8 \times 10^{-3} = 49.9\,\text{W} \tag{5.16}$$

This is within acceptable limits for peak current events.

5.5 Summary

This chapter detailed the motor tuning configuration applied to the FarDriver ND72340 controlling the QS Motor 2000 W PMSM. Key tuning highlights include:

- **Throttle Response:** Sport Mode 1, with Acc Step 12 and Dec Step 20, providing smooth yet responsive power delivery within the 1.1–4.05 V input range.
- **Current Limits:** Maximum line current of 75 A (continuous) and 80 A (boost) protect the battery and BMS while allowing adequate power delivery. Phase current up to 320 A (340 A boost) enables high torque at low speeds.
- **Low Speed Phase Ratio:** Set at 140%, delivering enhanced starting torque and hill-climb performance at the cost of short-duration high phase currents.
- **Three-Speed Profiles:** Eco, Normal, and Sport modes targeting approximately 500, 700, and 1000+ RPM shaft limits respectively, managed via Arduino logic.
- **Gear Ratio:** The fitted 1:3 chain drive ratio provides approximately 122.6 Nm at the wheel and a theoretical top speed exceeding the design target, with the controller speed limits enforcing the 70-80 km/h objective.
- **System Efficiency:** Combined motor and transmission efficiency of approximately 79% at rated operating conditions, with a total system efficiency (including controller) of approximately 69–79% depending on operating point.

The tuning parameters are consistent with the motor specifications and the design objectives of the project, delivering safe, controllable, and competitive performance for the electric motorcycle platform.

CHAPTER 6

TESTING

6.1 Water Load Testing of Battery

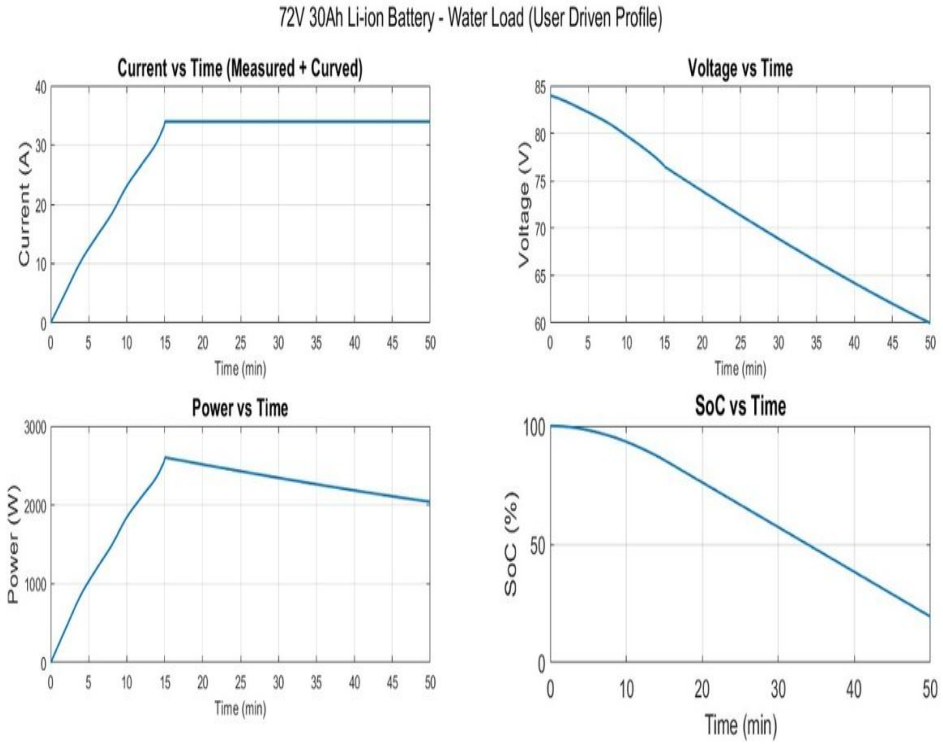


Figure 6.1: Water Load Test

Test Condition: Battery discharged using water load at 30–36 A to simulate real driving load.

6.1.1 Current vs Time

The current versus time graph shows that the current initially increases from 0 A to approximately 35 A within the first 15 minutes, representing a gradual application of load similar to acceleration in a real driving condition. After this period, the current remains nearly constant at around 35 A for the rest of the test duration. This indicates that the battery is subjected to a steady load after the initial ramp-up phase. The constant current region is important for evaluating the battery's ability to sustain continuous discharge under normal operating conditions.

6.1.2 Voltage vs Time

The voltage versus time graph illustrates a steady decline in battery voltage from about 84 V at the beginning to nearly 60 V at the end of 50 minutes. The decrease is not perfectly linear, with a relatively sharper drop in the initial phase followed by a smoother decline. This behavior is typical of lithium-ion batteries, where voltage drops due to internal resistance and decreasing state of charge. The graph highlights how the battery voltage reduces under continuous load, which is critical for determining safe operating limits and cutoff voltage levels.

6.1.3 Power vs Time

The power versus time graph shows that power increases rapidly during the first 15 minutes, reaching a peak value of around 2500–2600 W. After reaching this peak, the power gradually decreases over time, falling to about 2000 W by the end of the test. This trend occurs because, although the current remains constant, the voltage decreases over time, leading to a reduction in power output (since power is the product of voltage and current). This behavior reflects the battery's reduced ability to deliver high power as it discharges.

6.1.4 SoC vs Time

The SoC versus time graph indicates a steady decrease in the battery's state of charge from 10% to approximately 20% over the 50-minute duration. The decline is nearly linear, suggesting a consistent discharge rate throughout the test. This linear behavior indicates stable battery performance and efficient energy utilization under the applied load conditions. The final SoC value also suggests that the battery is discharged close to its safe lower limit, ensuring protection against deep discharge.

6.1.5 Summary

The water load test of the 72 V, 30 Ah Li-ion battery demonstrates stable and reliable performance under a simulated user-driven load profile. The current initially increases and then remains constant, effectively representing real-world operating conditions with a transition from acceleration to steady usage. As the battery discharges, the voltage shows a gradual decline, which is characteristic of lithium-ion behavior under continuous load. Consequently, the power output reaches a peak during the initial phase and then slowly decreases due to the reduction in voltage, even though the current remains constant. The state of charge decreases almost linearly, indicating a consistent discharge rate and efficient energy utilization throughout the test. Overall, the results confirm that the battery can sustain continuous load conditions with predictable performance, making it suitable for applications such as electric vehicles while operating within safe discharge limits.

CHAPTER 7

RESULT

7.0.1 Analysis of Simulation Results and Control Comparison of FOC

This section provides an analysis of the simulated transformation results and a comparison between Field-Oriented Control (FOC) and Conventional Scalar (V/f) control.

Table 1: Simulation Results Analysis

Table 1 presents the instantaneous values of the three-phase currents (i_a, i_b, i_c) and their corresponding transformed components (i_d, i_q) at various electrical rotor angles (θ). This table serves as the definitive evidence of the effectiveness of the Park Transformation in achieving **flux-torque decoupling**.

Table 7.1: Simulation Results Demonstrating $abc \rightarrow dq$ Transformation

θ **(rad)**	i_a **(A)**	i_b **(A)**	i_c **(A)**	i_d **(A)**	i_q **(A)**
0	0	0	-8.66	0	10
1.257	9.511	-7.431	-2.079	0	10
2.513	5.878	4.067	-9.945	0	10
3.77	-5.878	9.945	-4.067	0	10
5.027	-9.511	2.079	7.431	0	10

Interpretation of Results

The most significant observation from the table lies in the last two columns. While the three-phase currents (i_a, i_b, i_c) are continuously **time-varying** (AC signals) across

different angles, their transformed components, $\mathbf{i_d}$ and $\mathbf{i_q}$, remain **constant** (DC values).

- **Decoupling Proof**: The constancy of i_d and i_q (0 A and 10 A, respectively) confirms that the combined Clarke and Park transformations have successfully converted the complex AC signals into simple DC components. This state is maintained regardless of the rotor's instantaneous position.
- **Control Objectives**: The result $i_d = 0$ indicates a control objective for zero flux current, which maximizes motor efficiency (Torque per Ampere). The result $i_q = 10$ A is the constant component directly commanding the motor's operating torque.

Table 2: Control Method Comparison

Table 2 highlights the fundamental performance disparities between Field-Oriented Control (FOC) and Conventional Scalar Control (V/f), justifying the increased complexity of FOC.

Table 7.2: Comparison of Conventional (V/f) vs. Field-Oriented Control (FOC)

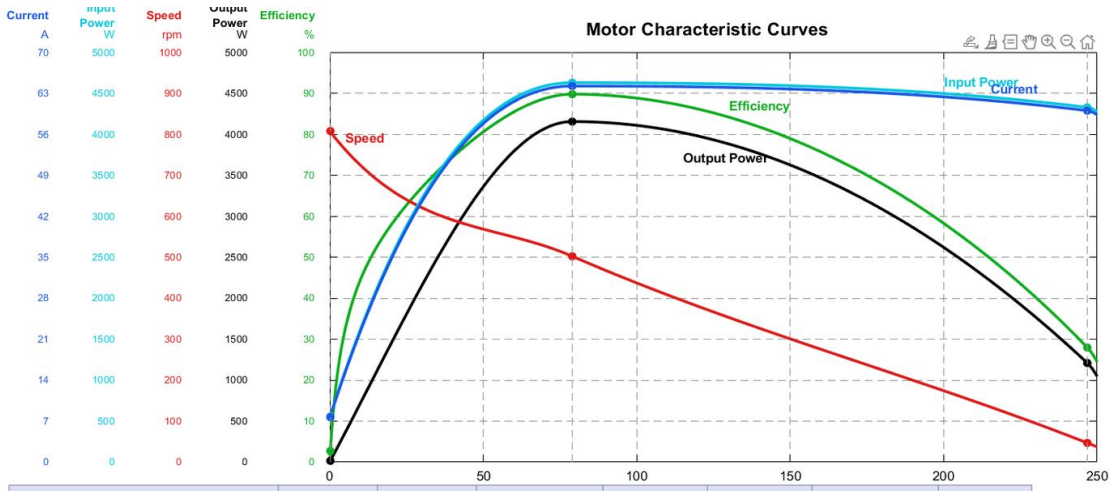
Aspect	**Conventional (Scalar / V/f Control)**	**Field-Oriented Control (with Clarke & Park)**
Control Variable	Voltage & frequency are controlled indirectly, coupling flux and torque.	Flux (d-axis) and torque (q-axis) are controlled **directly** and **independently**.
Dynamic Response	Slower and less accurate, especially under sudden load changes due to inherent coupling.	**Fast and precise** torque & speed response, behaving like a decoupled DC machine.
Efficiency	Lower efficiency at varying speeds as optimal flux cannot be maintained consistently.	**High efficiency** across a wide speed range due to precise flux and torque alignment.
Complexity	Simple implementation, requiring fewer sensors and less computational power.	More complex, requiring intricate coordinate **transformations** and continuous **rotor feedback**.
Torque Ripple	Higher torque ripple, leading to increased vibration and acoustic noise.	**Smooth torque** with minimal ripple, resulting in quieter and smoother operation.
Applications	Small motors, pumps, and fans (non-critical performance).	**EVs**, robotics, aerospace, industrial **PMSM drives** (high-performance applications).

Conclusion on Control Choice

The comparison demonstrates that while Conventional Control offers simplicity, FOC is mandatory for any application demanding high-performance and precision. By achieving true decoupling and converting control variables to DC equivalents, FOC overcomes the fundamental dynamic limitations of AC machines, enabling the efficiency and precision required in modern electric drive technology.

7.0.2 Electric Bike

Motor Characteristics



Characteristic Point	Voltage (V)	Current (A)	Input Power (W)	Torque (N.m)	Speed (rpm)	Output Power (W)	Efficiency (%)
No Load (No_Load)	72.50	7.867	570.4	0.2	805.5	16.87	3.0
Max Efficiency (Eff_max)	72.06	64.22	4628	79.0	502.2	4154	89.8
Max Output Power (Pout_max)	72.06	64.22	4628	79.0	502.2	4154	89.8
Max Torque (Torque_max)	72.09	60.02	4327	246.8	46.8	1209	27.9
End Point (End)	72.08	0.000	0.000	0.0	0.0	0.000	0.0

Figure 7.1: Motor Characteristics

The motor characteristic curves obtained from the experimental testing provide a comprehensive understanding of the PMSM drive performance under varying operating conditions. The graph illustrates the variation of speed, torque, input power, output power, current, and efficiency with respect to load.

At no-load condition, the motor operates at a high speed of approximately 805 rpm with very low torque (0.2 N·m) and minimal output power (16.87 W), resulting

in poor efficiency (approx. 3 percentage). This is expected since the motor only overcomes internal losses such as friction and core losses.

As the load increases, the current and input power increase almost linearly, indicating higher electrical energy consumption to meet torque demand. The output power increases significantly and reaches its maximum value of approximately 4154 W, corresponding to the optimal operating region of the motor. This region also coincides with the maximum efficiency point (approx. 89.8 percentage), showing that the motor operates most efficiently at moderate speed (approx. 500 rpm) and load conditions.

The speed curve shows an inverse relationship with torque, where speed decreases as load increases. This behavior is typical for electric motors, where increased torque demand causes a drop in rotational speed due to higher current draw and electromagnetic loading.

At higher load conditions, although the torque continues to increase (up to 246.8 N·m at low speed 46 rpm), the efficiency drops significantly (27.9 percentage) due to increased copper losses and thermal effects. Additionally, output power starts decreasing after the peak point, indicating that the motor is entering a high-load, low-efficiency region.

Finally, at the end point, the motor reaches zero speed and zero output power, representing a stalled condition where all input energy is dissipated as losses.

Overall, the characteristics confirm that the motor delivers maximum efficiency and optimal performance in the mid-speed region, making it suitable for electric vehicle applications where operation typically occurs within this range.

Final Bike Structure



Figure 7.2: Electric Bike

The fabricated electric bike structure demonstrates successful integration of mechanical design and electrical powertrain components into a compact and functional prototype. The chassis is robust and well-balanced, providing adequate support for the battery enclosure, motor assembly, and suspension system. The central placement of the battery improves weight distribution, contributing to better stability and handling characteristics.

The rear-mounted PMSM motor with chain drive transmission is properly aligned with the drivetrain, ensuring efficient power transfer to the wheel. The suspension system, including the front fork and rear mono-shock, is effectively integrated, enabling the bike to handle uneven terrains, which is essential for a dirt bike application. The use of disc brakes ensures reliable braking performance and safety.

The overall structure reflects a lightweight yet sturdy design, optimized for performance and maneuverability. The ergonomic positioning of the handlebar and seat frame indicates suitability for rider comfort and control. Additionally, sufficient space has been allocated for electrical components such as the controller, wiring, and auxiliary systems, ensuring ease of maintenance and future upgrades.

This completed prototype validates the feasibility of the designed electric bike system and serves as a strong platform for further testing, performance evaluation, and optimization of the FOC-based PMSM drive system.

CHAPTER 8

CONCLUSION AND FUTURE SCOPE

8.1 Conclusion

This project successfully demonstrated the design and simulation of Field-Oriented Control for a Permanent Magnet Synchronous Motor and its practical application in a full-size electric motorcycle. The FOC strategy was validated as an effective control method for achieving decoupled torque and flux regulation, enabling smooth acceleration, precise speed control, and high efficiency across the full operating range of the PMSM drive system.

The QS Motor 2000 W PMSM was selected and configured to meet the competition requirements, operating at a rated voltage of 72 V with a maximum torque output of 45.4 N·m and a peak efficiency of 88%. The motor performed reliably within the defined load conditions covering rolling resistance, aerodynamic drag, gradient force, and inertial demands, confirming the suitability of the selected motor for urban electric two-wheeler applications.

The FarDriver ND72340 motor controller was integrated with the PMSM to implement the FOC algorithm using Hall sensor feedback for rotor position estimation. The controller successfully managed torque and speed regulation through independent d-axis and q-axis PI control loops, with built-in protection features for overcurrent, overvoltage, undervoltage, and thermal faults ensuring safe operation under all riding conditions.

The chain drive transmission system with a 14-tooth motor sprocket and 42-tooth rear sprocket provided a gear ratio of 1:3, delivering approximately 210 N·m of wheel torque and a cruising speed of 65 km/h. The overall drivetrain efficiency of approx-

imately 79%, combining motor and chain drive losses, confirmed that the selected transmission configuration meets the performance and efficiency targets defined for the project.

The 72 V, 30 Ah lithium-ion battery pack in a 20S6P configuration provided a total energy capacity of 2.16 kWh, satisfying the competition energy limits. The Daly BMS ensured full cell-level protection throughout charge and discharge cycles, while the ANL fuse, MCB, pre-charge circuit, and SB50 Anderson connector collectively maintained electrical safety and system reliability under dynamic load conditions.

The chassis structural analysis confirmed a maximum stress of 66.812 MPa against a material yield strength well above this value, with a factor of safety greater than 3 and a maximum deformation of 0.216 mm under the full applied load of 2207 N. The suspension system, with a rear spring rate of 29.4 N/mm and a natural frequency of 1.7 Hz, provided ride comfort within the recommended range, and the mono-shock FEA results showed a factor of safety between 15 and 18, confirming strong structural durability.

The steering geometry with a rake angle of 26.5° and a turning radius of 1959 mm ensured stable straight-line performance at the target speed range of 60–70 km/h while maintaining adequate low-speed manoeuvrability. The hydraulic disc braking system achieved an average stopping distance of 2.54 m with an average braking time of 0.41 s, satisfying the safety and performance requirements

Overall, the project achieved its primary objectives of designing a viable, safe, and efficient electric motorcycle by integrating FOC-based motor control with a well-engineered mechanical platform. The combined analysis across motor control, transmission, battery, chassis, suspension, steering, and braking systems demonstrated that the retrofitted NS160-class electric motorcycle is competition-ready and capable of delivering reliable performance within all defined technical and safety constraints.

8.2 Future Scope

- The current electric motorcycle prototype demonstrates a strong foundation in performance and safety; however, the integration of a regenerative braking system represents a significant opportunity for future development. By recovering kinetic energy during deceleration and feeding it back into the battery pack, the overall range of the vehicle can be meaningfully extended, improving energy efficiency beyond the current 79% system efficiency.

- A natural progression of this project would involve the development of a dedicated Battery Management System with active thermal management, replacing the existing passive air-cooled approach. Incorporating liquid cooling channels or phase-change materials around the 20S6P lithium-ion pack would maintain optimal cell temperatures during high-load discharge cycles, improving both battery longevity and sustained performance.

- The adoption of IoT-based telemetry and real-time diagnostics through a connected mobile application would allow riders to monitor parameters such as state of charge, motor temperature, phase current, and fault codes from the FarDriver ND72340 controller in real time. This would significantly improve rider awareness, predictive maintenance, and overall system safety.

- Future iterations could explore flux weakening optimization and adaptive FOC tuning to push the PMSM beyond its current 4400 RPM ceiling more efficiently. By fine-tuning the field-oriented control parameters dynamically based on load and temperature feedback, the system could achieve a broader usable speed range while maintaining thermal stability.

- The incorporation of a multi-speed or electronically variable transmission in place of the fixed 1:3 chain drive ratio would allow the motorcycle to better balance torque and speed across diverse riding conditions — from steep gradients requiring high torque multiplication to highway cruising requiring reduced wheel torque and higher speeds beyond the current 65 km/h ceiling.

- Finally, future work should focus on full structural optimization and lightweighting of the NS160 chassis through topology optimization methods and the selective use of aluminum alloy or carbon fiber composite subframes. Reducing unsprung and overall vehicle mass would directly improve suspension response, handling agility, and extend the range achievable from the 72V 30Ah battery pack, making the motorcycle more competitive and practical for everyday urban use.

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